Fine-tuned language models learn tacit scientific judgment from institutional traces

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Abstract

Artificial intelligence matches or exceeds human performance on tasks with verifiable answers — from protein folding to Olympiad mathematics. Yet the capacity that governs scientific advance — judging which research ideas deserve pursuit before results are known — remains beyond AI's reach. This evaluative judgment is tacit, particularly in the social sciences: experts exercise it but cannot articulate it, and peer reviewers agree at barely above chance. Here we show that fine-tuning language models on institutional traces — decades of publication decisions in organisational psychology and management — recovers evaluative judgment inaccessible to both frontier AI and human expertise. Eleven frontier models including GPT-5.2 High, Claude Opus 4.6, and Gemini 3.1 Pro average 31.1% accuracy (chance = 25%), unable to distinguish good from bad research better than chance. Journal editors and editorial board members reach 41.6%, outperforming frontier AI but far from ideal. Our fine-tuned ensemble achieves 55.8%, surpassing all frontier models and expert panels by a large margin, while exhibiting calibrated self-knowledge — reliably distinguishing correct from incorrect predictions — a capacity absent in every other evaluator class. The missing piece in AI for science was not model scale but the right training signal — institutional traces learnable in as little as a single GPU-hour.

Introduction

Science is built on verifiable facts, but it is governed by unverifiable judgments. Artificial intelligence now matches or exceeds human performance on tasks with objectively verifiable answers—from protein structure prediction[1] to Olympiad-level mathematics[2] to competitive programming[3]. Yet the scientific enterprise depends equally on a different cognitive capacity: *evaluative judgment*—deciding which research ideas deserve pursuit before any experiment is run. As AI systems generate hypotheses and manuscripts at unprecedented scale[4, 5], the bottleneck in science has shifted from production to evaluation[6]. Whether AI can acquire this form of judgment—subjective, contextual, resistant to explicit codification—defines the next critical boundary in machine intelligence, where capability drops sharply from domains with clear answers to those requiring taste and discrimination[7].

Evaluative judgment is a paradigmatic instance of tacit knowledge: "we can know more than we can tell"[8]. Decades of effort to codify what makes a research idea good—structured review rubrics, editorial frameworks distinguishing originality from utility[9, 10]—have not produced agreement. A meta-analysis of 48 studies (19,443 manuscripts) found that reviewers agree at barely above chance levels[11]; in grant review, where proposals are evaluated before execution, agreement approaches zero[12]. Neither formal training, academic rank[13], nor editorial experience[14] reliably predicts review quality; review performance actually declines with experience[15]. Ethnographic work reveals that academic judgment operates through intuitive assessment and disciplinary sensibility rather than rule application[16]. Novel ideas—precisely the ones that matter most—generate the greatest disagreement[17], and the system systematically undervalues its most impactful work[18].

The quality signal is real: it manifests in the long-run sorting of publications across prestige tiers[19]. But the signal is irreducibly tacit—not in Polanyi's original sense of individual embodied skill, but in what Collins terms *collective* tacit knowledge: knowledge embedded in institutional practices that no individual participant can articulate, yet that the system reliably enacts[8, 20, 21]. Explicit evaluative criteria do not improve inter-rater reliability—the knowledge resists codification even when codification is attempted[11]—yet the institutional system nonetheless produces consistent quality stratification over time. This prolonged operation has left a trace: decades of publication decisions across prestige tiers constitute an institutional record of collective evaluative consensus that has never been exploited as a training signal for artificial intelligence.

This tacit character exposes a deep asymmetry in artificial intelligence. In domains with verifiable outputs, AI systems achieve superhuman performance[1, 2, 3]; but this advantage vanishes when the task shifts from verification to evaluation. When deployed in peer review—a setting where more than half of researchers already report using AI[22, 23]—large language models consistently inflate scores, recommend acceptance at rates far exceeding human panels, and systematically overlook novelty, the dimension that most distinguishes important from incremental work[24, 25, 26]. They evaluate how research is presented rather than what it contributes[27]. This leniency is not a prompting failure but a structural consequence of training: reinforcement learning from human preferences instils agreement-seeking behaviour that intensifies with further optimization[28, 29], producing models that default to approval rather than discrimination. The largest AI systems are trained to be agreeable, not discerning.

Here we construct a held-out benchmark of 120 article-derived research pitches in organizational psychology and management, balanced across four quality tiers and evaluated in 2,914 human ratings from 48 expert gatekeepers and 174 junior researchers. We test 18 AI systems spanning frontier reasoning models, chat models, and supervised fine-tuned models under an expert-derived prompt optimized to favor frontier performance. Frontier models perform only marginally above chance (31.1% versus 25%). Human evaluators perform somewhat better and preserve more balanced tier discrimination, but their judgments remain strikingly noisy, with inter-rater agreement near zero. Neither prompting the most powerful AI systems nor assembling expert panels solves the evaluation problem.

We align AI to these *institutional traces*. Individual reviewers disagree substantially, yet the institutional system, operating through iterated consensus over decades, produces reliable quality stratification that individual judgments do not[19]. It is this system-level signal, not individual reviewer accuracy, that supervised learning can exploit[20]. By applying supervised fine-tuning (SFT)—further training compact language models on labelled examples—to four base models using historical publication outcomes, we align them to implicit evaluative criteria that neither written instructions nor the largest AI architectures have successfully transmitted.

The results are stark. Our fine-tuned two-model ensemble achieves 55.8% accuracy—capturing approximately five times the headroom between chance and perfect performance compared to frontier models, and nearly double that of the expert majority vote, at a total training cost below \$300—dominated by a single proprietary API job; the open-source checkpoints required as little as one A100 GPU-hour. When all four fine-tuned models agree on a tier assignment—35.8% of articles—accuracy reaches 72.1%, providing a high-confidence triage signal unavailable to either frontier models or human panels. This echoes a broader pattern in AI development: the bottleneck is often not scale but alignment target, and targeted training on the right signal can outperform systems orders of magnitude more expensive[31]. Cross-architecture replication confirms the effect is not model-specific: all four base architectures show substantial gains after fine-tuning, with each single model exceeding both the frontier mean and the best frontier model. Beyond accuracy, the fine-tuned models exhibit calibrated self-knowledge: they reliably distinguish their own correct from incorrect predictions, a selective-prediction capacity absent in both frontier models and human experts. The model knows when to trust its own judgment.

These findings carry implications for the acceleration of science. Evaluative judgment is the binding constraint on scientific discovery, most acutely in the social and behavioral sciences, where hypotheses resist formal verification and submission volumes far outstrip expert bandwidth—precisely the fields where frontier AI fails most completely and where scalable, calibrated evaluation would yield the greatest returns. The mechanism should generalize to other domains with socially judged outcomes and weak ex ante verification—including creative industries, venture investing, grant allocation, hiring and promotion, and policy execution—whenever institutional decision histories accumulate large records of what later proved successful[19, 32]. In our benchmark, frontier-model performance clusters close to chance across the full breadth of proprietary and open-weight architectures, yet a compact fine-tuned system trained on accumulated editorial decisions exceeds both the most powerful AI systems and the expert panels whose collective decisions generated the training data. The barrier to AI evaluation was never raw capability. It was the absence of the right training signal.

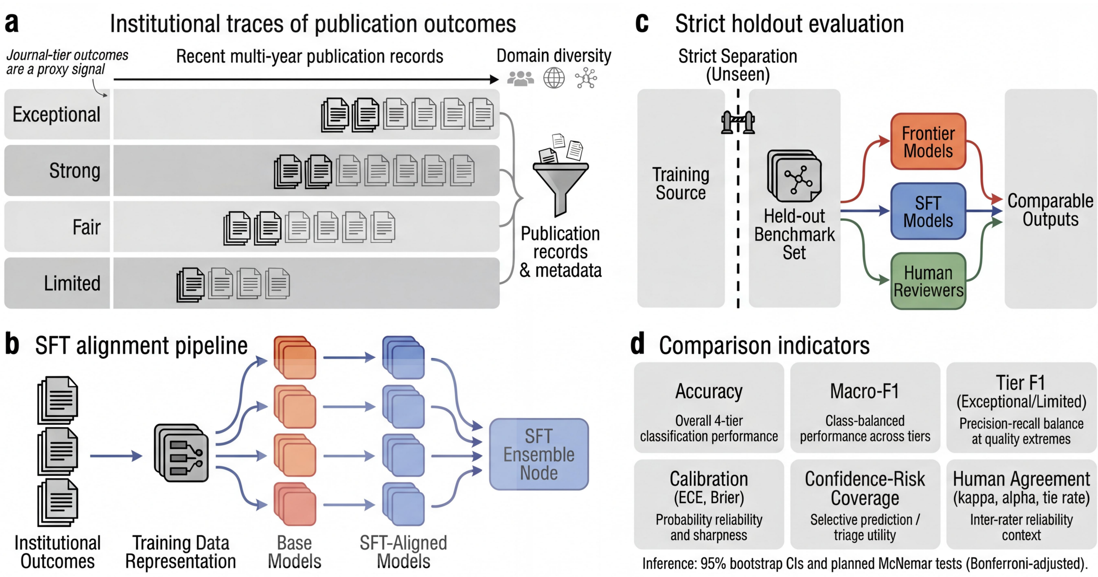
Results

A benchmark for evaluative judgment

Peer review rests on a cognitive capacity that has resisted formal specification: the ability to make early-stage significance judgments before full empirical execution, when journals, funders, and labs decide where to invest scarce attention and resources. To isolate this capacity, we constructed a held-out benchmark in organizational psychology and management comprising 120 article-derived research pitches, balanced across four quality tiers and spanning 15 research domains and 17 journals (see Methods for construction details). The source articles were all published after June 30, 2025. Each source article was reduced to a research pitch—a structured description of the core research question and theoretical framing—stripped of methodological detail, results, and publication identifiers. Articles were assigned to four quality tiers based on their publication outlet (exceptional, strong, fair, limited; 30 per tier; chance baseline 25%; Fig. 1).

Evaluators were asked to assign each pitch to one of the four tiers using an expert-derived assessment framework centred on originality and utility[9, 10]. The full evaluator landscape comprised eleven frontier reasoning models in a conservative cohort (including Gemini 3.1 Pro with contamination-risk caveat), three chat variants, four supervised fine-tuned (SFT) models with architecture-matched base controls, 48 expert gatekeepers, and 174 junior researchers—generating 2,914 independent ratings. The evaluation prompt was selected as the formulation that maximised frontier model performance, ensuring that any advantage observed for fine-tuned models represents a conservative estimate.

Figure 1 | Institutional traces and strict holdout evaluation framework



a, Journal-tier publication outcomes (Exceptional, Strong, Fair, Limited) define the institutional supervision signal and illustrate how long-run publication records serve as a proxy trace of collective gatekeeping judgment. **b**, These outcomes are converted into a standardized training representation that aligns multiple base models and supports ensemble construction. **c**, Training and held-out evaluation are strictly separated: frontier models, supervised fine-tuned models, and human raters are all evaluated on the same unseen benchmark under a common four-tier output mapping. **d**, The comparison framework specifies the metrics used throughout the manuscript, including accuracy, macro-F1, boundary-tier F1, calibration metrics (ECE and Brier score), selective prediction, and agreement diagnostics.

Frontier models and the anatomy of prediction collapse

If evaluative judgment were reducible to the kind of reasoning that frontier language models excel at—synthesising information, recognising patterns in text, applying criteria to cases—then the most capable models available should perform well above chance on this benchmark, particularly when given maximal advantage. The evaluation prompt was not a naive instruction: it was systematically stress-tested across prompt variants (Extended Data Fig. 1), incorporating expert-derived assessment criteria grounded in the editorial frameworks that define quality in the social sciences[9, 10]—originality, utility, and their interaction—with behavioural anchors calibrated to the collective consensus of the field's gatekeeping institutions. This prompt represents a deliberate attempt to transmit human evaluative knowledge to AI through explicit instruction, delivered to the most capable reasoning systems available. We also ran an auxiliary pairwise discrimination evaluation to test out-of-distribution ordinal judgment (Extended Data Fig. 2; Supplementary Table ST4), reported in detail below.

It is not enough. Across a conservative cohort of 11 frontier models—including GPT-5.2 High, Claude Opus 4.6, Gemini 2.5 Pro, Gemini 3.1 Pro, Qwen 3.5 Plus, Grok 4.1 Fast, and leading open-weight systems such as Kimi K2.5, DeepSeek V3.2 and GLM-5—mean accuracy was 31.1%, barely exceeding the 25% chance baseline and capturing only 8.1% of the available headroom between chance and perfect accuracy (Fig. 2a; see Methods for statistical tests). The failure runs deeper than accuracy alone: mean macro F1—a measure of balanced classification quality that penalises strategies ignoring minority tiers—was 0.236, below the 0.25 expected from random guessing. Models achieve above-chance accuracy on the tiers they over-predict while producing zero recall on tiers they ignore, a strategy that inflates accuracy but destroys balanced discrimination.

The failure follows three distinct patterns of prediction collapse that reveal how post-training alignment shapes model behaviour on evaluative tasks (Fig. 2b). The first pattern is strong-ceiling collapse: Claude Opus 4.6 assigns 87% of articles to the "strong" tier, producing a macro F1 of just 0.145—predicting virtually everything as above-average quality, using only two of the four available categories. The second is leniency-to-middle collapse: Seed 2.0 and Grok 4.1 heavily over-assign above-average tiers. The third is middle-tier clustering: GPT-5.2 High and DeepSeek V3.2 concentrate almost all non-tied predictions in strong/fair, compressing the distribution into a narrow band. Across the cohort, 6 of 11 models never predict "limited" for any article, producing zero recall on the lowest quality tier (Fig. 2c; full confusion matrices for all 11 models in Extended Data Fig. 3a; class-level distribution diagnostics in Supplementary Fig. 1). These patterns are consistent with reinforcement learning from human feedback optimising outputs toward agreeable, broadly acceptable responses[28, 29]—a property that serves conversational fluency but systematically undermines the discriminating judgments that scientific gatekeeping demands.

A contamination-risk caveat merits brief note. Gemini 3.1 Pro reached the highest frontier accuracy (38.8% avg8 article-mean), but its reasoning traces occasionally showed built-in search/tool-use behavior and partial reproduction of benchmark article content published only weeks before evaluation. We therefore retain Gemini 3.1 in the conservative benchmark with explicit caveating rather than treating it as a clean zero-shot-only comparator. Even under this conservative setup, explicit knowledge transfer remains insufficient: no frontier model reliably separates from the cohort after multiple-testing correction, and evaluative judgment is not recovered through prompting alone.

Figure 2 | Frontier models show structured prediction collapse

a, Eleven frontier models are compared with paired accuracy and macro-F1 bars under the conservative primary protocol, with Gemini 3.1 retained but explicitly marked for contamination risk. Accuracy bars show 95% binomial confidence intervals (n = 120 per model), while macro-F1 shows bootstrap 95% confidence intervals from tie-excluded majority-vote predictions. The main point is that the full frontier cohort remains close to chance despite using the best-performing prompt formulation. **b**, Predicted-tier distributions (100% stacked) show heavy concentration in the strong and fair tiers and sparse use of the limited tier, making the collapse pattern visible beyond scalar accuracy. **c**, Row-normalized confusion matrices for four representative flagships show distinct failure modes: Gemini 3.1 as the best-performing but still compressed model, GPT-5.2 High with middle-tier clustering, Claude Opus 4.6 with strong-tier ceiling behavior, and Qwen 3.5 Plus as the best open-weight reference. Extended collapse diagnostics across all 11 models are shown in Supplementary Fig. 4.

Human expertise and the limits of domain knowledge

The frontier models' failure might reflect a limitation specific to artificial systems—perhaps evaluative judgment requires the kind of tacit knowledge that only years of immersion in a research community can provide[8]. To test this, we recruited 48 expert gatekeepers: current editors and editorial board members at leading journals in organisational behaviour and management, individuals whose professional role consists precisely in making the quality judgments this benchmark demands.

Expert reviewers did outperform frontier models, but not by the margin their credentials would predict. Individual expert accuracy averaged 36.2%, significantly above chance but reflecting enormous variability—some experts performed well below chance while others in one case reached 100% on their rated articles (Fig. 3a; see Methods for tests of significance). When expert ratings were aggregated through majority vote, accuracy rose to 41.6% on the 89 articles that produced a clear plurality (31 of 120 articles yielded tied votes and were excluded), capturing 22.1% of the available headroom (Fig. 3b). Crucially, experts achieved macro F1 values of 0.353 (individual average) and 0.404 (majority vote), while the 11-model frontier average remained 0.236—below the 0.25 random baseline—demonstrating that humans genuinely discriminate across quality tiers rather than collapsing predictions into a narrow band. The accuracy gap between experts and frontier models (5.1 percentage points) understates the qualitative difference in evaluation behaviour.

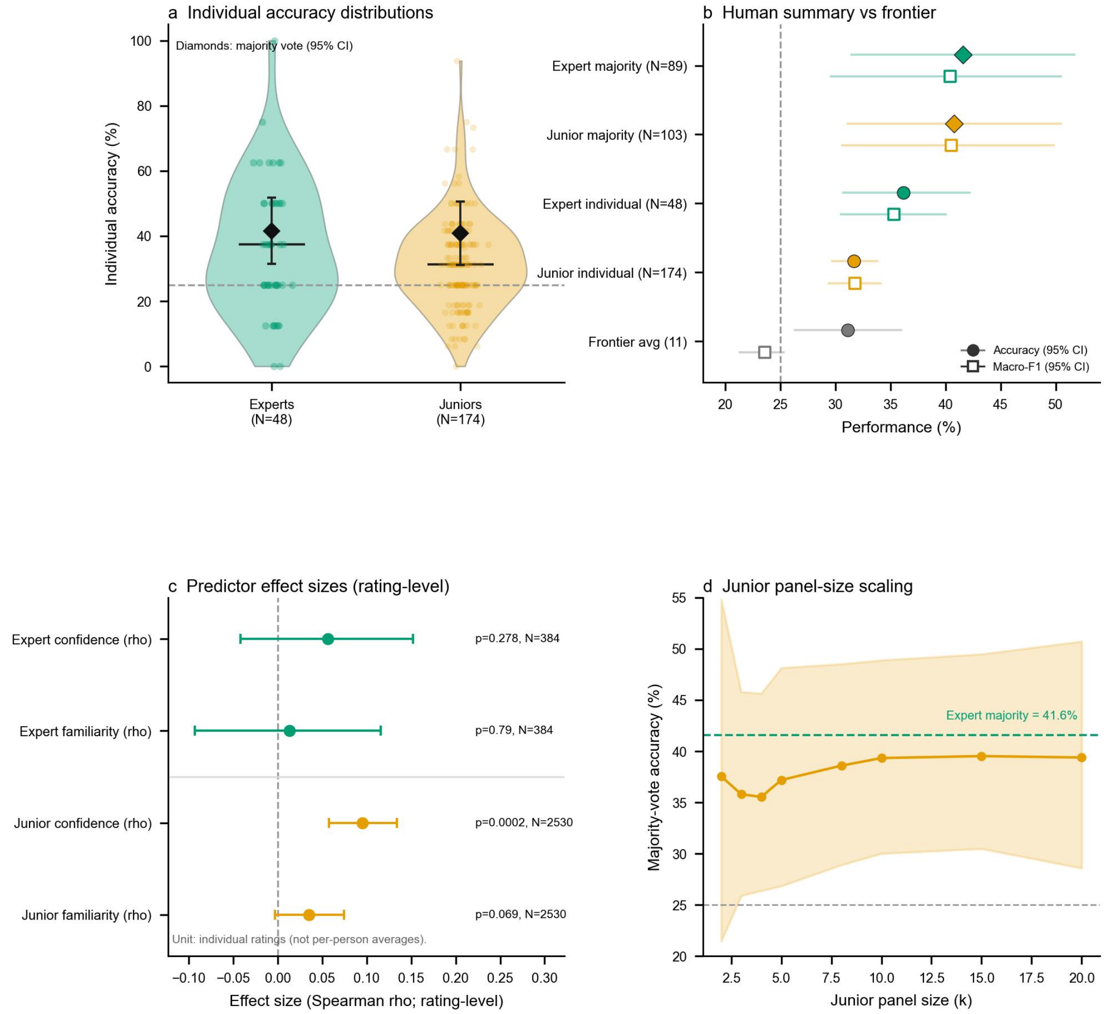
The pattern of agreement among experts exposes a telling dissociation. Categorical agreement—whether experts assign articles to the same tier—was near-chance (Fleiss' kappa = 0.047), barely distinguishable from noise[11]. Yet ordinal agreement—whether experts rank articles in roughly the same order of quality—was moderate (Krippendorff's alpha = 0.307). This gap suggests that experts share a coarse sense of which articles are better or worse but disagree substantially on where to draw categorical boundaries, consistent with evidence that expert judgment is dominated by noise that persists regardless of training or experience[30].

We searched systematically for expertise markers that might predict which reviewers perform well. No expertise marker predicted accuracy—not career stage, self-reported confidence, nor topic familiarity (Fig. 3c). These findings are consistent with a broader literature showing that neither domain expertise[33], career stage, nor publication record[13] reliably predicts evaluation accuracy.

Junior researchers—174 doctoral and postdoctoral scholars—performed comparably. Individual accuracy averaged 31.7%, and full majority vote reached 40.8% on 103 non-tied articles. Subsampling analysis—repeatedly drawing expert-sized panels from the junior researcher pool—yielded accuracy not significantly different from expert majority vote (Fig. 3d). Panel-size analysis revealed that accuracy plateaus at approximately 10 reviewers per article (Extended Data Fig. 4d), indicating a structural ceiling on what aggregation alone can achieve.

These results establish that evaluative judgment is genuinely difficult—not merely for AI systems optimised for conversational agreeableness, but for the human experts whose professional authority rests on this very capacity. The quality signal exists: institutional publication outcomes confirm it. But neither the most capable reasoning systems in artificial intelligence nor decades of accumulated domain expertise can reliably recover that signal from research pitches alone. The question is whether a different kind of learning—one that absorbs the implicit patterns in historical publication decisions rather than reasoning over explicit criteria—might succeed where both frontier AI and human expertise fall short.

Figure 3 | Human expertise helps, but noise remains dominant



a, Individual expert and junior accuracy distributions are shown with majority-vote reference markers, highlighting that experts outperform juniors on average but both groups remain highly heterogeneous; Supplementary Fig. 2 and Supplementary Table ST12 provide the full expert distribution details. **b**, Human reference configurations are compared against the frontier-model baseline on accuracy and macro-F1 with 95% uncertainty intervals, showing that human raters preserve more balanced discrimination than the frontier cohort. **c**, Confidence and topic familiarity are summarized as rating-level Spearman effect estimates on correctness, with bootstrap confidence intervals and permutation-based P values; the panel makes clear that these self-reported expertise markers explain little of the observed variation in accuracy. **d**, Monte Carlo matched-N junior subsampling shows how majority-vote accuracy changes with panel size, revealing early gains followed by a plateau rather than unlimited improvement; Supplementary Fig. 3 and Supplementary Table ST13 provide the corresponding resampling details.

Supervised fine-tuning on institutional traces recovers tacit judgment

We hypothesized that the information needed to discriminate quality tiers is not absent from language models' latent representations but rather inaccessible under standard prompting and reasoning regimes. To test this, we applied supervised fine-tuning (SFT) to four base models using a corpus of approximately 4,000 historical research articles labelled by journal tier—the institutional traces that encode the field's accumulated gatekeeping consensus[19]. The 120 benchmark articles were held out entirely and never seen during training. SFT does not inject new factual knowledge; it aligns latent representations to the distributional regularities through which institutional quality manifests in research pitches (see Methods for training details).

Before fine-tuning, we evaluated each base model on the benchmark to establish architecture-matched controls. Three of four base models scored at or near chance level (24.2–25.0%); GPT-4.1 reached 32.5%, modestly above chance but far below its fine-tuned counterpart (Extended Data Table 1). These results confirm that raw model capability alone, regardless of parameter count or architecture, cannot extract evaluative signal from research descriptions (Extended Data Fig. 3b).

Fine-tuning produced a marked improvement. All four models achieved 47.5–52.5% accuracy, gains of 15 to 28 percentage points over their respective base models (Fig. 4a). That these gains emerge consistently across two model families and two parameter scales confirms that the effect is not an artifact of a particular architecture, scale, or training implementation.

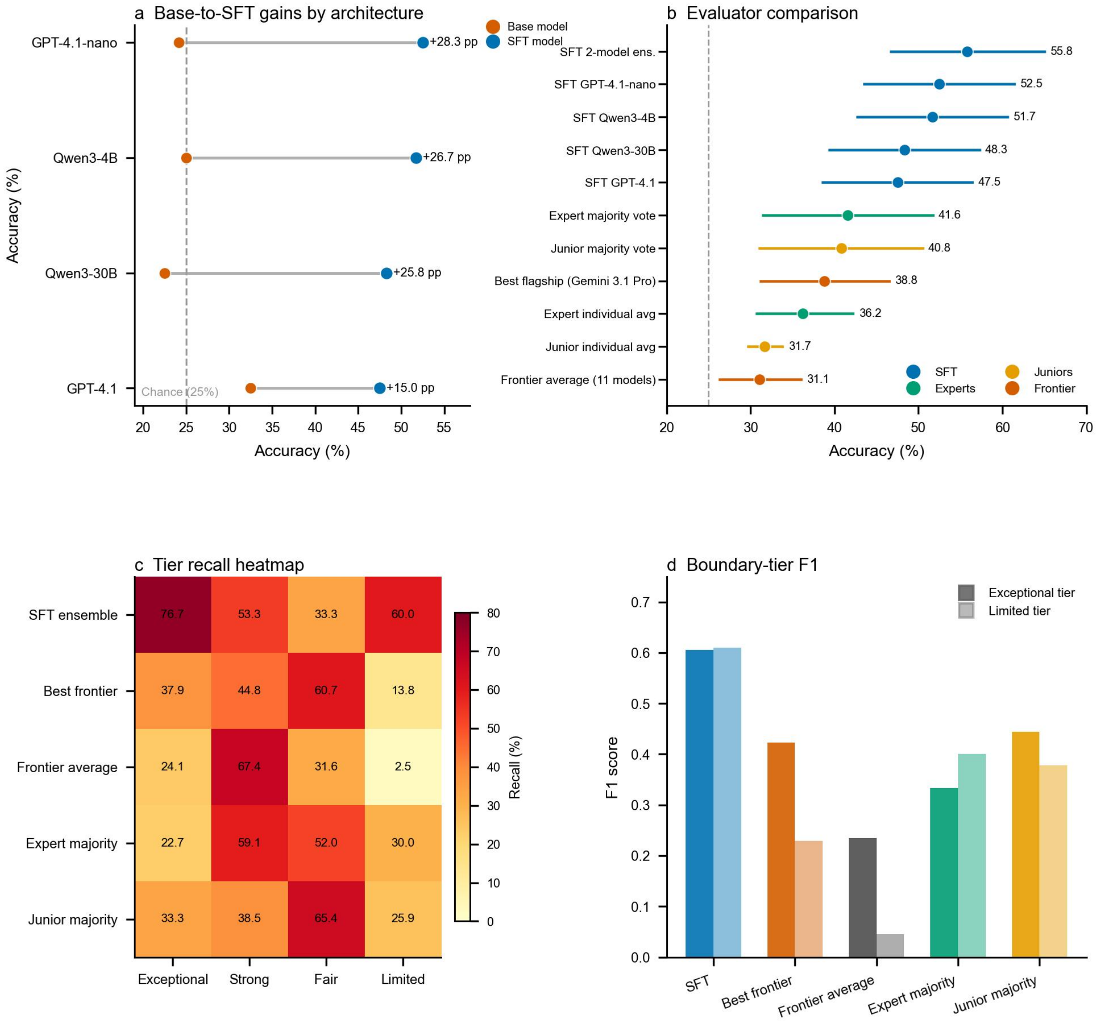
We systematically evaluated all six pairwise ensemble combinations among the four SFT checkpoints via probability averaging (Supplementary Table ST7). Accuracies ranged from 47.5% to 55.8%; all six combinations exceeded the frontier average (31.1%) by 16.4 to 24.7 percentage points, confirming that the ensemble advantage is robust to model pairing rather than a product of a single fortuitous combination (Extended Data Fig. 3c). The best-performing pair—GPT-4.1-nano (SFT) and Qwen3-30B (SFT), pairing a closed-source small model with an open-source mixture-of-experts architecture—achieved 55.8% (95% CI: 46.7–65.0%), capturing 41.1% of the chance-to-perfect headroom. The frontier average captures 8.1%; expert majority vote captures 22.1%. The SFT ensemble captures approximately five times the headroom of frontier models and nearly twice that of expert panels, with a macro F1 of 0.550 reflecting balanced discrimination across all four tiers (Supplementary Table ST6).

Comparisons confirmed the robustness of these differences. Against the frontier average (31.1%), the SFT ensemble outperformed by +24.8 percentage points (exact binomial, $p = 1.70 \times 10^{-8}$); against the best-performing frontier model under conservative protocol (Gemini 3.1 Pro, 39.1% majority accuracy on paired subset), by +15.7 percentage points (McNemar test, $p = 0.023$; Supplementary Table ST11). Because no individual frontier model significantly separates from the cohort after multiple-testing correction, the comparison against the frontier mean remains the most representative. Against individual evaluators, the ensemble surpassed the typical expert by a large margin (55.8% vs. 36.2% mean, $t(47) = -6.62$, $p < 0.001$, $d = 0.96$), with 85% of experts and 94% of junior researchers scoring below the ensemble (Fig. 4b). Against collective majority votes, SFT showed advantages of 13.5 percentage points over expert majority ($p = 0.112$, reflecting the reduced power of majority-vote comparison on 89 non-tied articles) and 15.5 percentage points over junior majority ($p = 0.0375$).

The per-tier analysis reveals where the advantage concentrates and exposes a fundamental distinction in evaluation style (Fig. 4c,d). SFT achieved substantially higher balanced detection rates (F1) than expert panels at both quality extremes—exceptional-tier F1 = 0.61 versus 0.33 for experts (nearly double), limited-tier F1 = 0.61 versus 0.40 (over 50% higher). Human panels are conservative gatekeepers—high precision when they commit to an extreme judgment (63% for exceptional, 60% for limited), but missing the vast majority of articles at each quality end (23% recall for exceptional, 30% for limited). Experts concentrated 80% of their predictions in the middle tiers. SFT functions as a sensitive detector, predicting across the full quality spectrum with a distribution closer to the true base rate (77% recall for exceptional at 50% precision; 60% recall for limited at 62% precision). Its false positives at the exceptional tier were predominantly near-misses—articles from the adjacent strong tier, not catastrophic misclassifications. In any system where the cost of missing exceptional work exceeds the cost of additional review—the typical condition when submission volumes far outstrip expert bandwidth—a sensitive detector addresses the more consequential failure mode.

Total training cost across all four models was under $300, dominated by a single proprietary fine-tuning job (~$200 for GPT-4.1); the two open-source checkpoints required approximately one and eight A100 GPU-hours respectively, with near-zero inference cost (Supplementary Table ST5). The critical ingredient is the alignment target—institutional traces—not model scale.

Figure 4 | SFT on institutional traces recovers tier discrimination



a, Architecture-matched base-to-SFT comparisons quantify gains attributable to supervised fine-tuning rather than architecture choice alone, showing that all four aligned models improve markedly over their own unfine-tuned baselines. **b**, A unified evaluator comparison places the SFT ensemble, single SFT models, human references, the best frontier model, and the frontier average on the same metric scale with confidence intervals, making the overall ranking transparent in one panel. **c**, Row-normalized recall across the four quality tiers compares the SFT ensemble, best frontier model, frontier average, expert majority vote, and junior majority vote, showing that SFT recovers the full tier structure rather than collapsing toward the middle. **d**, Boundary-tier F1 for the exceptional and limited tiers highlights where SFT gains are strongest and why they matter for gatekeeping decisions at the two extremes of article quality.

The mechanism: alignment target, not scale or reasoning

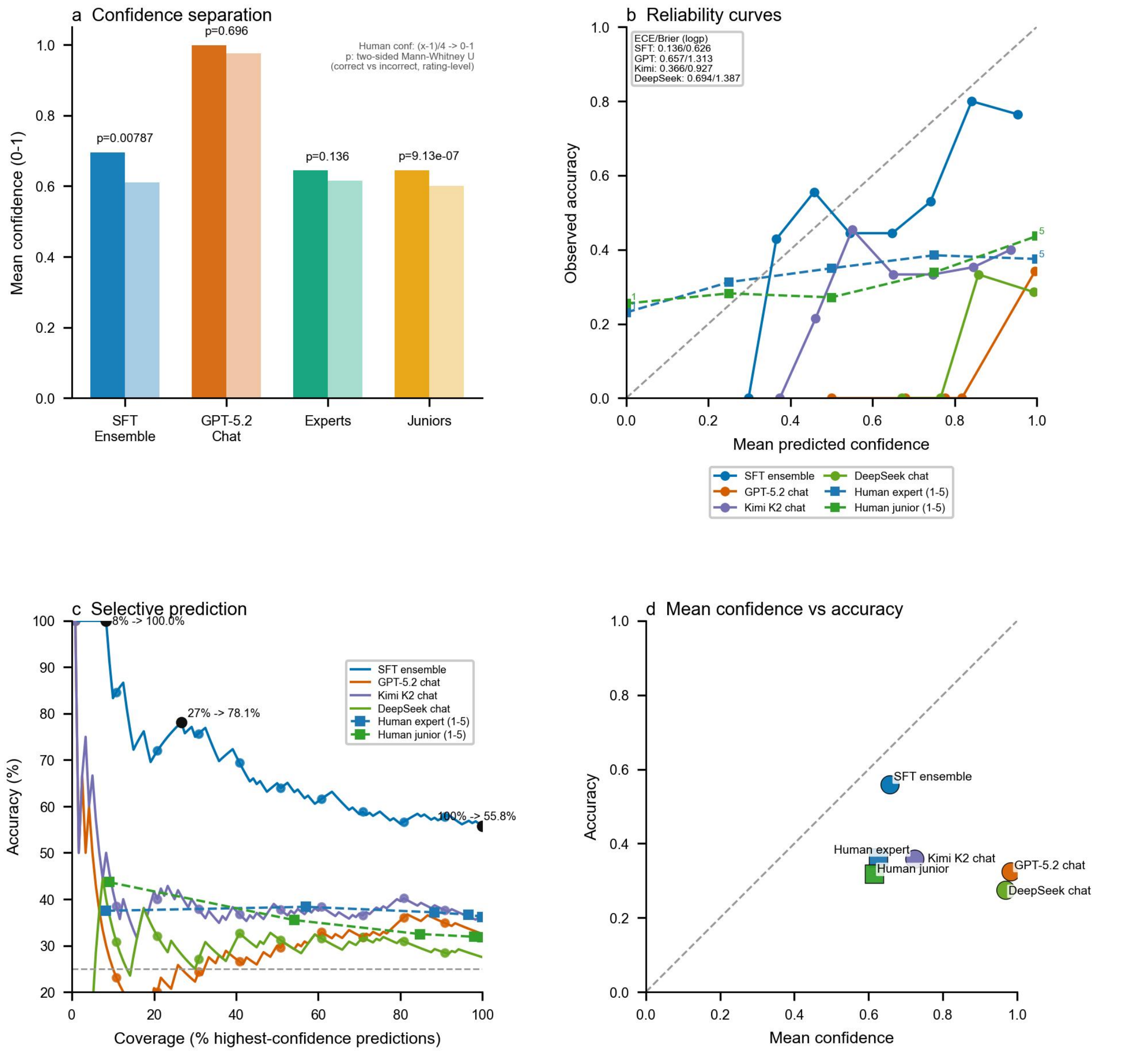
The SFT results pose a mechanistic puzzle. Why does a simple training procedure on journal-tier labels unlock evaluative performance that neither hundred-billion-parameter frontier models nor experienced human reviewers achieve? Three lines of evidence converge on a single answer: SFT aligns models to tacit evaluative regularities that resist extraction through explicit reasoning but yield to example-based distributional learning.

Calibrated metacognition. A defining property of genuine expertise is knowing what one knows—the capacity to assign higher confidence to judgments that prove correct. The SFT ensemble exhibited this property: its prediction confidence was significantly higher on correct predictions than on incorrect ones (confidence gap = +0.086, $p = 0.008$; expected calibration error = 0.136; Fig. 5a). Three of four individual SFT models showed the same pattern, with the fourth trending in the same direction, confirming that calibration is a consistent property of the fine-tuning approach, not an artifact of a single model.

This calibration stands in sharp contrast to both frontier AI and human evaluators (Extended Data Fig. 4a-c). GPT-5.2 High exhibited extreme, undifferentiated overconfidence: near-maximum confidence regardless of whether its predictions were correct, despite near-chance accuracy (ECE = 0.657, a measure of how well stated confidence matches actual accuracy; Fig. 5b). Human expert confidence was similarly non-discriminative. Junior researcher confidence showed a weak association with accuracy, but the effect was far too small for practical use. Neither frontier AI nor human evaluators can reliably identify which of their own judgments to trust (Fig. 5d).

The SFT models can. When predictions were restricted to the highest-confidence subset (approximately the top quarter), accuracy rose to 78.1%; at the top 8.3% (10 of 120 articles), accuracy reached 100%—every high-confidence prediction was correct (Fig. 5c). This selective prediction capacity—the ability to flag cases where the model's judgment is highly reliable while abstaining on uncertain ones—represents a form of metacognitive triage absent in every other evaluator class. In practice, this enables a confidence-based workflow: route high-confidence predictions directly and flag uncertain cases for human review.

Figure 5 | Calibration and selective triage under unified confidence scaling



a, Mean confidence for correct and incorrect predictions is compared for the SFT ensemble, GPT-5.2 chat, experts, and juniors. Human 1-5 confidence is mapped to 0-1 by (x - 1) / 4, and P values are one-sided Mann-Whitney U tests comparing whether correct ratings are more confident than incorrect ratings at the rating level. The panel asks whether each evaluator knows when it is right. **b**, Reliability curves compare mean predicted confidence with observed accuracy for the SFT ensemble, GPT-5.2 chat, Kimi K2 chat, and DeepSeek chat; human confidence-level calibration points are overlaid on the same 0-1 scale. ECE and Brier score are reported for probabilistic evaluators to summarize miscalibration and overall confidence quality. **c**, Selective-prediction curves show how accuracy changes as lower-confidence cases are deferred, with AI predictions ordered from highest to lowest confidence and human curves defined by confidence thresholds; this panel visualizes the practical triage value of calibrated abstention. **d**, Mean confidence is plotted against realized accuracy for each evaluator relative to the calibration diagonal, highlighting overconfidence in chat models and better alignment for the SFT ensemble.

SFT versus reinforcement learning. If evaluative judgment is tacit—integrated, holistic, and resistant to decomposition into explicit inferential steps[8]—then training methods requiring articulated reasoning chains should be less effective than methods that learn directly from examples. This prediction follows from a well-established principle in cognitive science: verbal articulation of holistic judgments systematically degrades their quality[35], and recent evidence confirms that chain-of-thought prompting yields negligible or negative gains on tasks outside mathematics and symbolic reasoning, with accuracy drops of up to 36 percentage points on tasks where human deliberation also hurts[36, 37]. We tested this by training reinforcement-learning variants on the same base architectures in their reasoning-enabled configurations (see Methods). In RL evaluation, run-pooled avg8 accuracy was 40.34% (384/952), article-mean avg8 was 40.25%, and majority-vote accuracy was 41.67% on non-tied articles (45/108; 12 ties). This is above the frontier average (31.1%) and slightly above the best frontier avg8 benchmark (Gemini 3.1 Pro, 38.8%) but remains below SFT single models (47.5–52.5%), matching the ordering in Extended Data Table 2 and Extended Data Fig. 5. Per-architecture RL checkpoints show the same ranking pattern. The pattern of errors is instructive: when the model's reasoning chain diverged from its final label, accuracy dropped markedly; when reasoning and label agreed, performance approached SFT levels. This suggests that explicit deliberation can override otherwise sound evaluative intuitions—not that reasoning is harmful per se, but that for tacit judgments, direct example-based alignment extracts the signal more reliably than reasoning-mediated optimization. Reinforcement learning from human feedback remains the standard approach for broad language model alignment[28, 34], but the epistemic structure of the target task determines which training method is most effective.

Pairwise head-to-head discrimination. To confirm that SFT develops generalized evaluative judgment rather than category-specific pattern matching, we tested models in a pairwise format entirely absent from training. Models received two articles simultaneously and judged which represented higher-quality research. The GPT model family provides a controlled comparison spanning three levels of capability: the pre-fine-tuning GPT-4.1 baseline, the frontier GPT-5.2 High, and our SFT GPT-4.1. Pairwise accuracy rose monotonically across this hierarchy: baseline GPT-4.1 reached 76.25% (183/240), GPT-5.2 High reached 82.08% (197/240), and SFT GPT-4.1 reached 90.83% (218/240; Extended Data Fig. 2; Supplementary Table ST4). Gemini 3.1 Pro, added as a conservative frontier comparator in this auxiliary pairwise set, reached 84.88% (203/239). Fine-tuning on institutional traces thus exceeded not only the model's own pre-training baseline but also frontier comparators with orders of magnitude more parameters and training compute. The advantage was largest on the hardest pairs—adjacent tiers separated by a single quality step: SFT achieved 87.50% versus 57.50% for the baseline, a 30-percentage-point gain precisely where fine-grained discrimination matters most. Cross-model comparison confirmed the pattern: SFT GPT-4.1 outperformed Gemini 3.1 Pro overall (90.83% vs. 84.88%) and on adjacent-tier pairs (87.50% vs. 72.15%). Accuracy increased monotonically with tier distance for all models, confirming that SFT's main advantage lies in resolving fine-grained quality differences rather than easy extreme contrasts.

Together, these three lines of evidence—calibrated self-knowledge, the SFT-over-RL advantage, and generalized pairwise discrimination—converge on a single conclusion: the evaluative signal is embedded in the distributional regularities of institutional traces and is most effectively recovered by direct example-based alignment rather than by explicit reasoning or increased model scale.

Voting consistency and aggregation asymmetry

If individual SFT models capture evaluative signal, a natural question is whether aggregating across models amplifies that signal further—and whether the same holds for frontier models and human panels. The final mechanism question is whether aggregation improves judgment through additional votes alone, or through the quality and diversity of the voters. Fig. 6 presents a unified comparison of consensus-filtering strategies across AI and human evaluator classes, revealing a fundamental asymmetry in how aggregation functions across evaluator types.

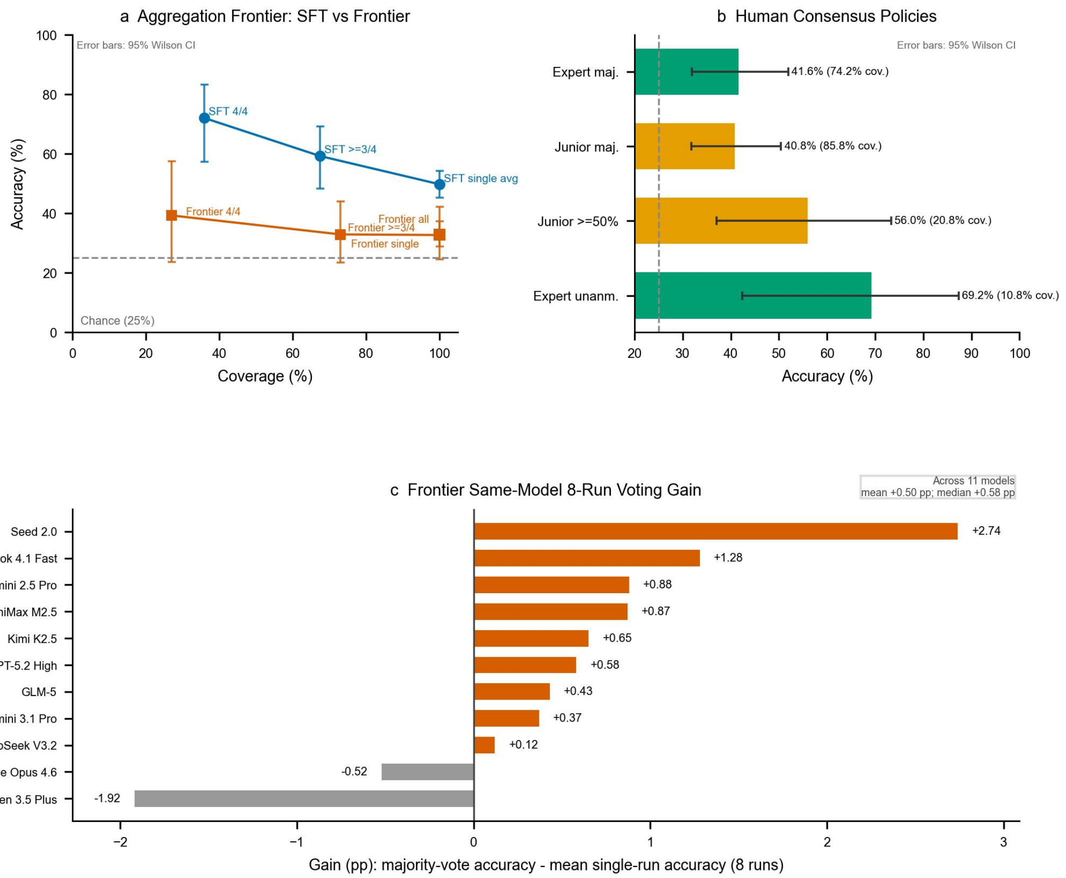
Across the four SFT models, full consensus (4/4) occurs on 35.8% of articles (N = 43) and reaches 72.1% accuracy—the highest precision achieved by any evaluator configuration in this study (Fig. 6a). Within this unanimous subset, accuracy is sharpest at the quality extremes: 94.1% for exceptional-tier and 80.0% for limited-tier articles, while middle tiers remain harder (strong 50.0%, fair 33.3%; Supplementary Table ST15). Relaxing to >=3/4 agreement expands coverage to 67.5% at 59.3% accuracy, still nearly 20 percentage points above the best majority-vote baseline. When SFT models split 2/2, accuracy falls to 30.8%, near chance—indicating that model disagreement is a reliable uncertainty signal, not random noise (Fig. 6a).

Human voting exhibits a steeper tradeoff. Expert unanimous agreement (among articles with at least two expert ratings) achieves 69.2% accuracy at 10.8% coverage, while >=50% student consensus reaches 53.6% accuracy at 23.3% coverage. Without consensus filtering, student and expert full-panel plurality accuracies are 40.0% and 39.2% on all 120 articles (Fig. 6b).

The contrast with frontier models is equally stark. For the 11 stochastic frontier models in the conservative cohort, repeated sampling produced only modest within-model gains: majority-vote accuracy exceeded avg8 by +0.50 percentage points on average (median +0.58 pp; range -1.92 to +2.74; Fig. 6c). Cross-model voting using a fixed four-model diverse set (Gemini 3.1 Pro, Claude Opus 4.6, GPT-5.2 High, GLM-5) fared no better. This set combines leading proprietary systems with strong open-weight coverage while maintaining model-family diversity. Plurality voting reached 32.7% on the 104 articles with complete model-level votes, and higher consensus thresholds did not recover performance (>=3/4: 32.9%; 4/4: 39.3% at low coverage; Fig. 6a). Frontier models fail to create meaningful consensus because their RLHF-induced biases cause correlated errors: different models collapse toward the same middle categories on the same articles.

The asymmetry between evaluator classes is instructive. Where frontier model aggregation produces no gain, SFT cross-model consensus delivers a 22-percentage-point boost (from ~50% single-model average to 72.1% at 4/4 consensus). This arises because SFT models, despite independent training on different architectures, converge on a shared evaluative signal (inter-model Cohen's kappa = 0.37–0.50, versus human inter-rater kappa approximately 0; Supplementary Table ST15). Their errors are sufficiently diverse across architectures that consensus effectively filters out model-specific noise. The practical implication is a selective-evaluation workflow: auto-route the 36% of submissions where all four SFT models agree (at 72% accuracy), flag the remainder for human expert review, and concentrate scarce reviewer attention on the genuinely ambiguous cases.

Figure 6 | Aggregation asymmetry after metric-definition alignment



a, Cross-model consensus policies are compared for the four SFT models and a fixed four-model frontier set. Accuracy intervals use Wilson 95% confidence intervals, and coverage indicates the share of articles satisfying each agreement rule. The contrast is substantive: stricter consensus improves SFT precision sharply, whereas frontier cross-model voting remains weak even at higher agreement thresholds. **b**, Human consensus policies compare expert majority, junior majority, junior >=50% vote share, and expert unanimity among articles with at least two expert ratings. Accuracy intervals in this panel also use Wilson 95% confidence intervals, and labels report the associated coverage, making the human accuracy-coverage tradeoff explicit. **c**, Same-model 8-run gains for frontier systems are summarized as majority-vote accuracy minus mean single-run accuracy, showing that repeated sampling alone yields only modest improvements and does not solve the underlying judgment problem.

Discussion

The central finding of this study is not that AI outperforms human experts at evaluative judgment—it is that the evaluative signal was never missing. Frontier models fail at scientific evaluation despite excelling at tasks of comparable linguistic complexity (Fig. 2). Forty-eight expert gatekeepers perform significantly above chance but capture barely a fifth of the available headroom (Fig. 3). Yet a simple fine-tuning procedure on historical publication outcomes transforms chance-level base models into evaluators that surpass both (Fig. 4a). The information needed to discriminate research quality was always present in language models' latent representations; what was missing was the alignment target that makes it accessible—not a better prompt, a larger model, or more inference-time reasoning, but the institutional trace that encodes a field's collective evaluative consensus[20].

This reframes a foundational question about AI and unverifiable judgment. The dominant strategy in AI development—scaling parameters, data, and inference-time computation—has produced extraordinary gains on tasks with objectively verifiable outputs[1, 2, 3]. For evaluative judgment, this strategy fails: frontier systems capture only 8.1% of the available headroom, with a mean macro F1 of 0.236—below the 0.25 expected from random guessing (Fig. 2a). This failure persists even when frontier models receive a systematically optimised evaluation prompt that encodes expert-derived assessment criteria and editorial frameworks—the most thorough attempt at explicit knowledge transfer that domain expertise can produce. The SFT ensemble, by contrast, captures 41.1% of the headroom with a macro F1 of 0.550 reflecting balanced discrimination across all four tiers (Fig. 4a), at a training cost below $300—with open-source models requiring as little as a single GPU-hour. The distinction maps onto Collins's taxonomy of tacit knowledge[20]: evaluative judgment is collective tacit knowledge—embedded in institutional practices that no individual participant can articulate—which yields to example-based distributional learning but resists extraction through explicit instruction. The dissociation between SFT and reinforcement learning is consistent with this interpretation: SFT aligns representations directly to examples and outperforms RL, whose explicit reasoning channel can override otherwise sound evaluative intuitions (Extended Data Table 2; Extended Data Fig. 5). For judgments that are genuine but tacit, the critical resource is not more computation but the right training signal, already deposited in the historical record of human institutions.

Three properties of the fine-tuned models define the practical path forward. First, calibrated metacognition: the SFT ensemble reliably distinguishes its own correct from incorrect predictions (Fig. 5a), enabling confidence-based triage. When predictions are restricted to the highest-confidence subset, accuracy reaches 100%—every high-confidence prediction is correct (Fig. 5c)—a self-knowledge absent in both frontier AI, which exhibits catastrophic miscalibration (Fig. 5b), and human experts, whose confidence bears no relationship to their accuracy. Second, cross-model consensus as a reliability signal: when all four independently trained SFT models agree on a tier assignment (35.8% of articles), accuracy reaches 72.1%—the highest precision achieved by any evaluator configuration in this study (Fig. 6a). Frontier model consensus, by contrast, produces no improvement because RLHF-induced biases generate correlated errors across architectures (Fig. 6c). Third, these properties jointly enable a selective-evaluation workflow: auto-route the third of submissions where AI consensus is unanimous, flag the remainder for expert review, and concentrate scarce reviewer attention on the genuinely ambiguous cases where human judgment is most needed.

Several limitations qualify these conclusions. Our benchmark draws from a single domain—organisational psychology and management—chosen because its evaluation criteria are among the most explicitly codified in the social sciences. Whether the institutional-trace mechanism transfers to disciplines with different epistemic cultures requires direct replication. Most fundamentally, institutional traces are a proxy for quality, not an objective ground truth. They encode the accumulated consensus of a gatekeeping system with well-documented biases—toward incremental over novel work[18], toward established methodologies, toward dominant paradigms. The SFT model learns what the system historically rewarded, which is not identical to what is objectively best. This limitation is real but bounded: in domains where quality is irreducibly unverifiable, the long-run institutional consensus is the best available approximation[19], and the model's calibrated uncertainty provides an internal check on its own reliability that the system it learned from lacks.

The mechanism we demonstrate—mining institutional traces to align AI to tacit human consensus—is not specific to peer review. In any domain where collective human evaluation has operated over time, the historical record of what was selected, funded, or rewarded constitutes a learnable signal: venture investing, grant allocation, creative industries, hiring and promotion[19, 32]. Expert panels routinely fail to predict outcomes in these domains, yet the institutional record of what ultimately succeeded encodes evaluative regularities that prospective judgment cannot access. In our benchmark, frontier-model performance clusters near chance across the full breadth of proprietary and open-weight architectures, yet a compact fine-tuned system trained on accumulated editorial decisions exceeds both the most powerful AI systems and the expert panels whose collective decisions generated the training data (Fig. 4b). The barrier to AI evaluation was never raw capability. It was the absence of the right training signal.

Author note

Qiuping Peng and Liyun Zhang assisted with data collection and project management.

Data availability

The de-identified benchmark data, model prediction files, human-rating files, figures, tables, and reproducibility scripts supporting this study will be released through the project repository at https://github.com/FutureTech-OB/tacit-scientific-judgment. The repository will also document the supervised fine-tuning dataset structure and release boundary.

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Methods

Study design. Evaluative judgment in science—the ability to assess which research ideas deserve investment before results are known—is widely recognized as tacit (difficult to articulate or teach) and institutionally distributed[8, 16, 20]: ethnographic evidence shows that even experienced gatekeepers rely on intuitive assessment and disciplinary sensibility rather than rule application[16], and decades of structured rubric design have not improved inter-rater reliability[11]. This study was designed to test whether AI systems can recover such judgment from institutional decision traces—the accumulated publication outcomes left behind by decades of editorial gatekeeping—when asked to assess research ideas before empirical results are known. We centered the protocol on early-stage research-pitch evaluation rather than full-paper critique, and imposed a shared four-tier decision space across all evaluator classes—human and AI—so that performance differences reflect differences in judgment rather than in task format, criterion framing, or access to stylistic cues.

Benchmark construction and rationale for labels. We constructed a balanced benchmark of 120 article-derived research pitches in organizational psychology and management (30 per tier; chance baseline 25%), each extracted from a peer-reviewed article sampled from the predefined 19-journal source universe described in Supplementary Methods SM5 and published after the first half of 2025. The four tiers—exceptional, strong, fair, and limited—were derived from journal-level publication outcomes mapped to a pre-specified tier framework and reviewed by domain experts (Supplementary Methods SM5). Journal-level labels were chosen because they represent the most stable institutional record of long-run gatekeeping decisions currently available at scale[19, 20]: individual reviewer judgments are dominated by noise[30], but the system-level sorting that produces prestige-tier differentiation integrates iterated editorial consensus over time. This makes the training target explicit: the model learns to recover collective institutional judgment, not to approximate any single reviewer's opinion.

A balanced design was chosen to prevent models from exploiting class-frequency priors (that is, learning to predict whichever tier is most common rather than actually evaluating quality) and to enable interpretable per-tier comparisons, especially at the quality extremes where editorial decisions are most consequential. Tier balance is confirmed in Supplementary Table ST2; domain coverage across 15 topic areas is reported in Supplementary Table ST3.

Input standardization and extraction workflow. Evaluator comparisons are meaningful only if all evaluators see the same information. Each source article was therefore transformed into a standardized research-pitch text—a structured summary presenting the core research question and theoretical framing without results, methods detail, journal identity, or author identity—using a fixed extraction workflow (Supplementary Methods SM4). The primary extractor was Qwen3-235B-A22B-Instruct, a large language model selected for extraction quality and format stability after cross-validation against alternative strong models. We generated five textual formulations—varying in the balance of theoretical context, research-question specificity, and methodological detail—and selected the research-question-with-context form as the primary representation through pilot evaluation on a held-out development set, because it preserves theoretical motivation while stripping later-stage evidence that would trivially encode publication outcomes. This normalization isolates idea-level assessment and reduces shortcut learning—the tendency for models to latch onto superficial cues such as writing quality or venue prestige rather than evaluating the underlying idea. The same policy was applied across all model families and to human-facing materials so that evaluator comparisons remained behaviourally aligned.

Training corpus and leakage control. The supervised training corpus contained 3,994 processed research-idea instances paired with tier labels, derived from source articles drawn from the same 19-journal source universe, and was fully disjoint from the 120 benchmark idea pitches. Tier labels followed the same journal-level mapping used in evaluation. Ambiguous samples were excluded during curation to reduce avoidable label noise in the training target.

Temporal separation was introduced to control contamination risk (the possibility that a model might have encountered benchmark source articles during its own training): source articles underlying the benchmark pitches were published after mid-2025, whereas the base models used for fine-tuning had pretraining knowledge cutoffs before that period. Frontier models released in 2026 were evaluated as zero-shot baselines (that is, without any task-specific training or examples). Because provider-level exposure risk can differ, we retained Gemini 3.1 in the conservative primary frontier benchmark with an explicit contamination-risk marker and corresponding caveat in interpretation.

Because publication outcomes are influenced by factors beyond idea quality—execution quality, writing quality, reviewer-assignment effects, editorial fit—and because the model input captures only idea-level information, a noise ceiling is expected by design[30] (Supplementary Methods SM8). This ceiling affects absolute accuracy interpretation while leaving relative comparisons internally valid, since the same information constraint is imposed across all evaluator classes.

Evaluation criteria and prompt-freezing strategy. All evaluators used the same two-axis criteria framework centered on originality and usefulness, consistent with editorial guidance literature in management research[9, 10]. A shared rubric—a standardized scoring guide describing what distinguishes each quality tier—was imposed on both humans and AI so that score differences would reflect evaluator behaviour rather than criterion mismatch. Tier definitions and prompt variants are provided in Supplementary Methods SM6.

Three prompt variants were pre-specified before final runs: an expert-anchored rubric (using the language and standards of experienced journal editors), a simplified plain-language rubric, and a journal-anchored rubric (referencing specific publication venues as quality anchors). The expert-anchored and journal-anchored variants yielded comparable frontier performance in pre-analysis checks, but journal-anchored prompts induced models to enumerate journal lists and hunt for venue-specific stylistic cues during reasoning; the expert-anchored variant was therefore frozen as the primary prompt to avoid this confound, making downstream comparisons conservative for frontier systems rather than optimized for fine-tuned models. Cross-model prompt-sensitivity analysis is reported in Extended Data Fig. 1. Evaluations were run in zero-shot mode to keep conditions comparable across model families and to avoid injecting exemplar-dependent anchors that can shift decisions independently of evaluator capability. Unresolved model outputs (cases where a model failed to produce a valid tier label) were coded as incorrect for fixed-denominator reporting (overall non-compliance rate <1% across 10,560 individual runs; 27 failures, predominantly from Gemini 3.1 Pro). Gemini-specific prompt-sensitivity results are reported in Supplementary Table ST1.

Human evaluation protocol. The human study was approved by institutional review board review (Project No. THU-04-2026-0034). Participants provided informed consent. Experts participated voluntarily without financial compensation.

Two panels were used to separate experienced gatekeeper judgment from trainee judgment under the same task definition. The expert panel comprised 48 raters (384 ratings; mean 3.2 ratings per pitch), recruited through personal professional networks to ensure high domain relevance and intrinsic motivation; two anonymous response identifiers were reconciled as two distinct raters. The junior panel comprised 174 doctoral and postdoctoral researchers (2,530 ratings; mean 21.1 ratings per pitch), recruited via doctoral and postdoctoral networks; junior raters who spent fewer than one minute per pitch were excluded to remove perfunctory responses. Experts each rated exactly eight assigned benchmark pitches (mean 8.0). Unfiltered expert data were prespecified as the primary expert analysis because filtering reduces pitch-level vote counts and increases tie sensitivity, limiting the pitches on which majority voting can be computed. Filtered sensitivity analyses are reported in Supplementary Table ST10.

For each benchmark pitch, raters reported prior exposure (yes/no), tier assignment (Top, Top-, Good, Fair mapped to exceptional, strong, fair, limited), confidence (5-point Likert scale), and topic familiarity (5-point Likert scale). The full survey instrument and procedural details are provided in Supplementary Methods SM7. We collected confidence and familiarity because prior work shows that peer-review agreement and evaluator quality are weakly coupled to conventional expertise signals[11, 12]; these variables allowed direct testing of whether the same pattern held in our dataset. Demographic and background summaries are in Supplementary Table ST8; individual expert accuracy values are in Supplementary Table ST12; matched-N Monte Carlo details are in Supplementary Table ST13; and a concise prior-exposure descriptive summary is reported in Supplementary Table ST14.

AI model families and inference protocols. We evaluated three AI families under one frozen instruction scaffold. The conservative frontier cohort included 11 reasoning models—Claude Opus 4.6, GPT-5.2 High, Gemini 2.5 Pro, Gemini 3.1 Pro, Qwen 3.5 Plus, DeepSeek V3.2, Seed 2.0, MiniMax M2.5, Kimi K2.5, Grok 4.1 Fast, and GLM-5. Gemini 3.1 Pro is retained with a contamination-risk caveat because reasoning traces sometimes exhibited built-in search/retrieval behavior. Each frontier model was sampled eight times per pitch to account for the inherent randomness in text generation. The primary frontier metric was pitch-mean avg8 accuracy (the average accuracy across all eight runs per pitch, then averaged across pitches). For discrete diagnostics, we used per-pitch majority vote from the same eight runs and reported effective sample size after tie exclusion. Protocol coverage is summarized in Supplementary Table ST16.

We additionally evaluated chat/log-probability tracks (GPT-5.2 chat, Kimi K2 chat, DeepSeek Chat) and architecture-matched base controls—the same underlying models before any fine-tuning, serving as baselines to isolate the effect of training from pre-existing capability. Each base model was evaluated on the full 120-pitch benchmark using the same frozen prompt and four-label log-probability classification (a method that extracts the model's internal confidence score for each of the four tier labels in a single pass, producing a probability distribution rather than a text response) as its SFT counterpart, providing architecture-matched controls that isolate the effect of fine-tuning from pre-existing model capability. Non-thinking models (base, chat, and supervised fine-tuned checkpoints) were evaluated by four-label token log-probability classification, which yields deterministic class probabilities and avoids free-text parsing failure modes. The same log-probability protocol was used for all four SFT checkpoints and their six pairwise probability-averaging ensembles (Supplementary Table ST7).

This dual protocol reflects a difference in what each API surface exposes: reasoning APIs produce stochastic natural-language completions best stabilized by repeated sampling, while non-reasoning APIs expose token-level log-probabilities that give calibrated label distributions in a single deterministic call—enabling both confidence analysis and direct probabilistic comparison. Cost and protocol comparisons are summarized in Supplementary Table ST5.

Supervised fine-tuning. Supervised fine-tuning (SFT) is a procedure in which a pre-trained language model is further trained on a curated dataset of input–output examples to specialize it for a particular task[39]. We fine-tuned four base models spanning two model families and multiple scales: GPT-4.1, GPT-4.1-nano (a smaller, more efficient variant), Qwen3-4B-Instruct, and Qwen3-30B-A3B-Instruct (a mixture-of-experts architecture with 30 billion total parameters but only 3 billion active at any time). The cross-family, cross-scale design tests whether the evaluative signal is recoverable from institutional traces generally or specific to a particular architecture, and whether compact models can match or exceed larger ones when given the right training target[34].

Each training example consisted of the frozen evaluation prompt wrapping a research-question-with-context pitch as input, with a single tier-label token (one of four: exceptional, strong, fair, limited) as the completion target. The same prompt template was used for training and evaluation, eliminating prompt-mismatch confounds. Given input text x and label $y \in \{1, 2, 3, 4\}$, training minimized label-token negative log-likelihood—a standard objective that increases the probability the model assigns to the correct tier label:

$$L\_NLL(\theta) = -(1/N)\ \Sigma\_\{i=1\}^N \log p\_\theta(y^(i) \mid x^(i))$$

with loss computed on label tokens only and input tokens masked from gradient updates. Input masking is critical: it prevents the model from memorizing prompt tokens and forces learning exclusively the mapping from research content to quality tier. The training corpus comprised 3,994 processed article-derived research-idea instances drawn from the 19-journal source universe, approximately balanced across tiers, with ambiguous source articles excluded. Total training cost was below $300 USD across all four models, with the largest single cost arising from GPT-4.1 API fine-tuning (~$200); GPT-4.1-nano required ~$10 via API, while the open-source Qwen3-4B and Qwen3-30B checkpoints required approximately one and eight A100 GPU-hours respectively (Supplementary Table ST5). Qwen checkpoints were trained locally using TRL (Transformer Reinforcement Learning, an open-source library for fine-tuning language models); GPT checkpoints were trained via the OpenAI fine-tuning API. Full hyperparameters and hardware are provided in Supplementary Methods SM1.

A two-model ensemble was built by averaging softmax-normalized label probabilities—a mathematical operation that converts raw model outputs into a probability distribution summing to 1—from GPT-4.1-nano (SFT) and Qwen3-30B-A3B (SFT), then selecting the maximum-probability class. We evaluated all six pairwise SFT ensembles and retained this pair as primary because it gave the highest held-out benchmark accuracy (Supplementary Table ST7). Because all six pairwise ensembles exceeded the frontier average by 16.4 to 24.7 percentage points, the core finding is robust to ensemble composition rather than dependent on a single fortuitous combination.

Architecture-matched base controls serve as the primary comparison for isolating the SFT effect: because each fine-tuned model is evaluated against its own pre-training checkpoint on the identical benchmark, observed gains are attributable to the fine-tuning procedure rather than to differences in base-model capability or architecture.

Reinforcement-learning ablation. Reinforcement learning (RL) is a training approach in which a model learns by generating outputs, receiving reward signals for better answers, and adjusting its behaviour accordingly—in contrast to supervised fine-tuning, which learns directly from labelled examples. To test whether explicit reasoning optimization improves this task beyond supervised alignment, we trained reasoning-enabled Qwen3-4B and Qwen3-32B checkpoints with a modified GRPO-style objective[28, 31, 38]—a variant of the Group Relative Policy Optimization algorithm used to train models that reason step-by-step before answering (Supplementary Methods SM2–SM3). The implementation removed the KL penalty (a regularization term that prevents the model from deviating too far from its starting behaviour), used token-level normalization, and adopted asymmetric clipping to favour positive-advantage updates (that is, the algorithm more readily reinforces successful strategies than it penalizes unsuccessful ones). We also introduced adaptive privileged sampling for low-accuracy items[40, 41] to maintain reward contrast during training.

RL checkpoints were evaluated using the same 8-run sampling protocol as frontier models, with run-pooled and pitch-mean accuracy reported alongside non-tied majority vote. RL training inherently requires reasoning-enabled model configurations that generate explicit chain-of-thought—a sequence of intermediate reasoning steps written out before the final answer—before a final label; this is not a confound but the mechanism under test. The comparison asks whether explicit deliberation aids or impedes evaluative accuracy—a question motivated by converging evidence from cognitive science that verbal articulation can degrade holistic judgment[35], and from machine learning that chain-of-thought prompting yields negligible or negative gains on tasks outside mathematics and symbolic reasoning[36, 37]. Reinforcement learning from human preferences has additionally been shown to instil agreement-seeking behaviour[28, 29] and reduce output diversity through mode collapse (the tendency for a model to converge on a narrow range of safe, agreeable outputs), and GRPO-style methods represent the current state of the art for reasoning optimization[31]. This RL track was therefore treated as a mechanism test rather than a deployment candidate: if explicit chain-of-thought policy optimization cannot match direct supervised alignment on this task, the implication is that the evaluative signal resides in the institutional traces themselves rather than in reasoning architecture.

Pairwise discrimination experiment. We designed a label-free pairwise task to distinguish gains in intrinsic discrimination from gains in rubric alignment. Each trial presented two research pitches sampled from different tiers, and the model selected the stronger one without seeing tier labels or rubric text—testing whether a model can tell which of two ideas is better even without the four-category framework. We used a fixed 240-pair stratified set (80 pairs each for tier distances 1, 2, and 3, where distance 1 means adjacent tiers and distance 3 means the most separated tiers).

Models evaluated were GPT-4.1 baseline, GPT-5.2 High, Gemini 3.1 Pro, and SFT GPT-4.1. Because this task removes absolute category assignment, improvements here are interpretable as improvements in relative quality discrimination rather than prompt-following alone. Results are reported in Extended Data Fig. 2 and Supplementary Table ST4.

Voting-consistency analysis. Consensus analyses were prespecified to quantify how reliability changes with stricter agreement thresholds across evaluator classes—in other words, what happens to accuracy when we only count cases where multiple evaluators agree. For SFT, we analyzed 4/4, 3/4, and 2/4 cross-model agreement policies. For humans, we analyzed junior vote-share thresholds and expert unanimity subsets. For frontier models, we analyzed both within-model repeated-sampling aggregation and fixed cross-model voting on a diverse top-four-model subset (Gemini 3.1 Pro, Claude Opus 4.6, GPT-5.2 High, GLM-5).

Inter-rater consistency used Fleiss' κ (a statistical measure of agreement among multiple raters, where 0 indicates chance-level agreement and 1 indicates perfect agreement) for human panels and pairwise Cohen's κ for model-model agreement. Majority-vote results excluded tied pitches and always reported effective non-tied N. Ties were not broken randomly because random resolution introduces avoidable variance and obscures whether an evaluator family is genuinely indecisive on difficult cases. Full agreement diagnostics are reported in Supplementary Table ST15.

Statistical analysis. All analyses were run in Python (NumPy, SciPy, pandas, statsmodels). The primary endpoint was four-class exact-match accuracy—the proportion of benchmark pitches for which an evaluator's tier assignment matched the ground-truth tier. Secondary endpoints included macro-F1 (the harmonic mean of precision and recall averaged equally across all four tiers, penalizing strategies that ignore minority classes), per-tier precision/recall/F1, confusion matrices (tables showing how predictions map onto true labels), and calibration diagnostics.

Ordinal inter-rater agreement was assessed with Krippendorff's alpha, an agreement coefficient that accounts for the ordinal structure of tier ratings, where values near 0 indicate chance agreement and 1 indicates perfect agreement. Paired evaluator comparisons used McNemar tests (a statistical test for comparing two classifiers on the same set of items by examining where they disagree) on pitch-level paired correctness vectors. Frontier-cohort heterogeneity was tested with Cochran's Q (a test for whether multiple classifiers perform equivalently across the same items). Ordinal or non-Gaussian analyses used Spearman correlation, Mann–Whitney U, and Kruskal–Wallis tests. Individual-level comparisons tested whether the SFT ensemble accuracy differed from the distribution of individual human evaluator accuracies using a one-sample t-test with the ensemble score as the reference value. Confidence intervals were estimated by bootstrap resampling (drawing 10,000 random samples with replacement from the data to estimate uncertainty). Multiple testing was handled with Bonferroni and Benjamini–Hochberg procedures (standard corrections that guard against false positives when many statistical tests are conducted simultaneously; Supplementary Table ST11).

The analysis plan defined three primary hypothesis comparisons: SFT ensemble versus expert majority vote, SFT ensemble versus best frontier model under the conservative protocol (Gemini 3.1 Pro), and SFT ensemble versus GPT-5.2 chat baseline. Additional pairwise tests were reported as secondary analyses with corrected P values. Headroom is defined as (accuracy - chance) / (100% - chance), where chance = 25%, giving the fraction of improvable performance captured by a given evaluator. Human judgment analyses incorporated the noise framework of Kahneman et al.[30] to interpret the dissociation between low categorical agreement and moderate ordinal agreement. Majority-vote comparisons involve reduced effective N because tied pitches are excluded, which limits statistical power relative to individual-level tests; this is particularly relevant for the SFT-versus-expert-majority comparison, where tie exclusion reduces the evaluable set.

Calibration for probabilistic evaluators used expected calibration error (ECE—a measure of how well a model's stated confidence matches its actual accuracy across different confidence levels) and Brier decomposition (a scoring rule that separates overall prediction quality into reliability, resolution, and uncertainty components); selective prediction was evaluated by plotting accuracy as a function of coverage under confidence-threshold sweeps (that is, how much accuracy improves when the model is allowed to abstain on its least confident predictions). Monte Carlo matched-N analyses (5,000 random draws) were used to compare junior and expert majority voting at equivalent panel sizes, drawing expert-sized panels from the junior researcher pool and computing majority-vote accuracy on each draw. Label normalization across source-article metadata, human surveys, and model outputs used deterministic mapping rules (Supplementary Table ST9).

Data and code availability. All preprocessing rules, prompts, model inventories, hyperparameters, and sensitivity analyses are documented in Supplementary Methods and Tables. The benchmark split was fixed before final model comparisons, and label normalization rules were deterministic across source-article metadata, human surveys, and model outputs. Analysis code, processed benchmark files, prompts, and table-generation scripts are prepared for public release with the manuscript package. Human data will be de-identified before release in accordance with approved ethics procedures.

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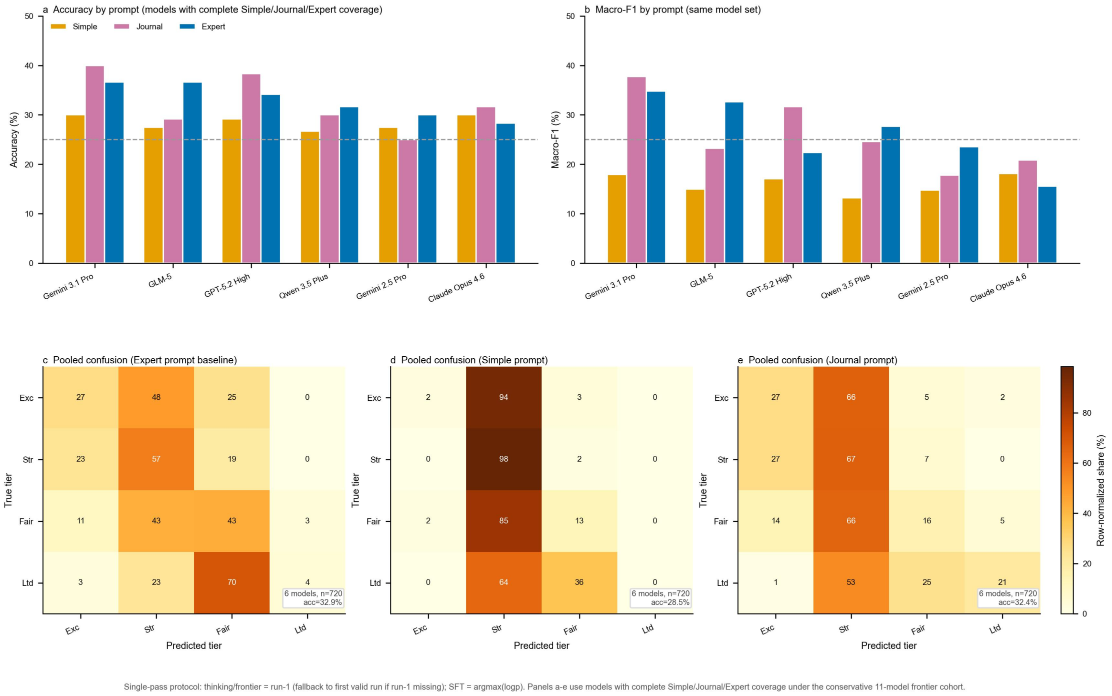
Extended Data

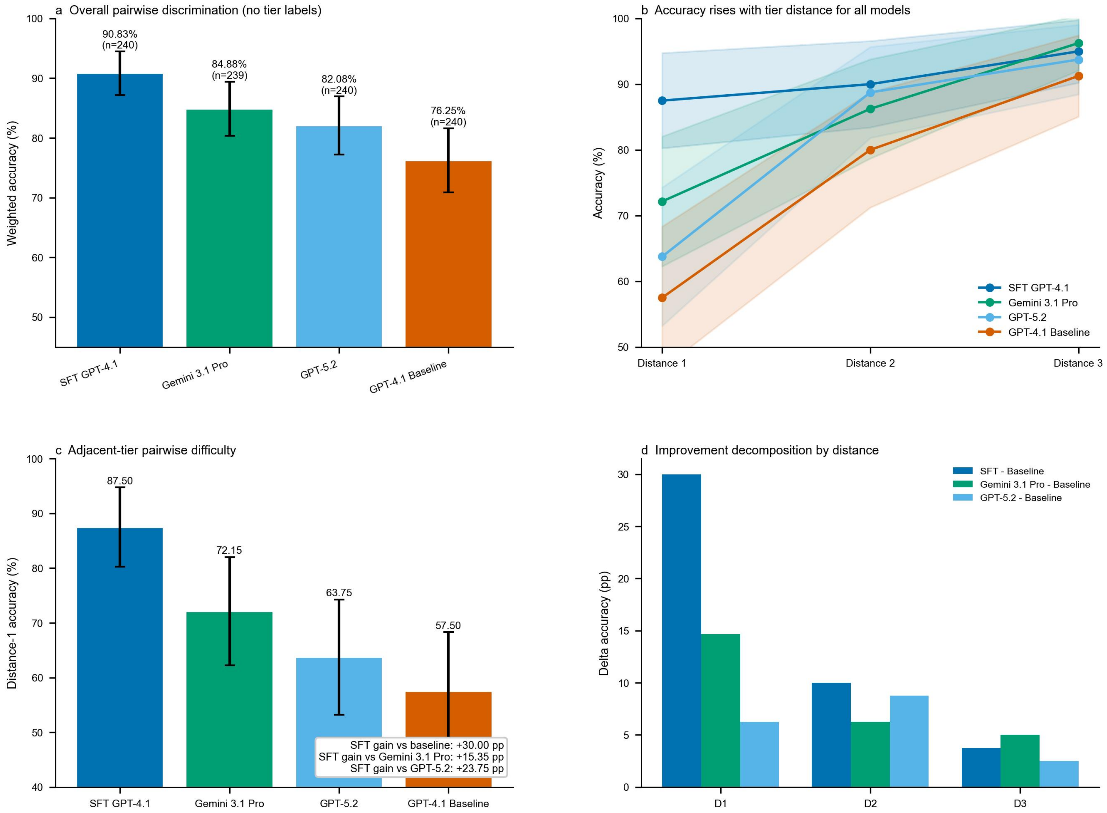
Extended Data Figures

Extended Data Figure 1 | Cross-model prompt-sensitivity landscape



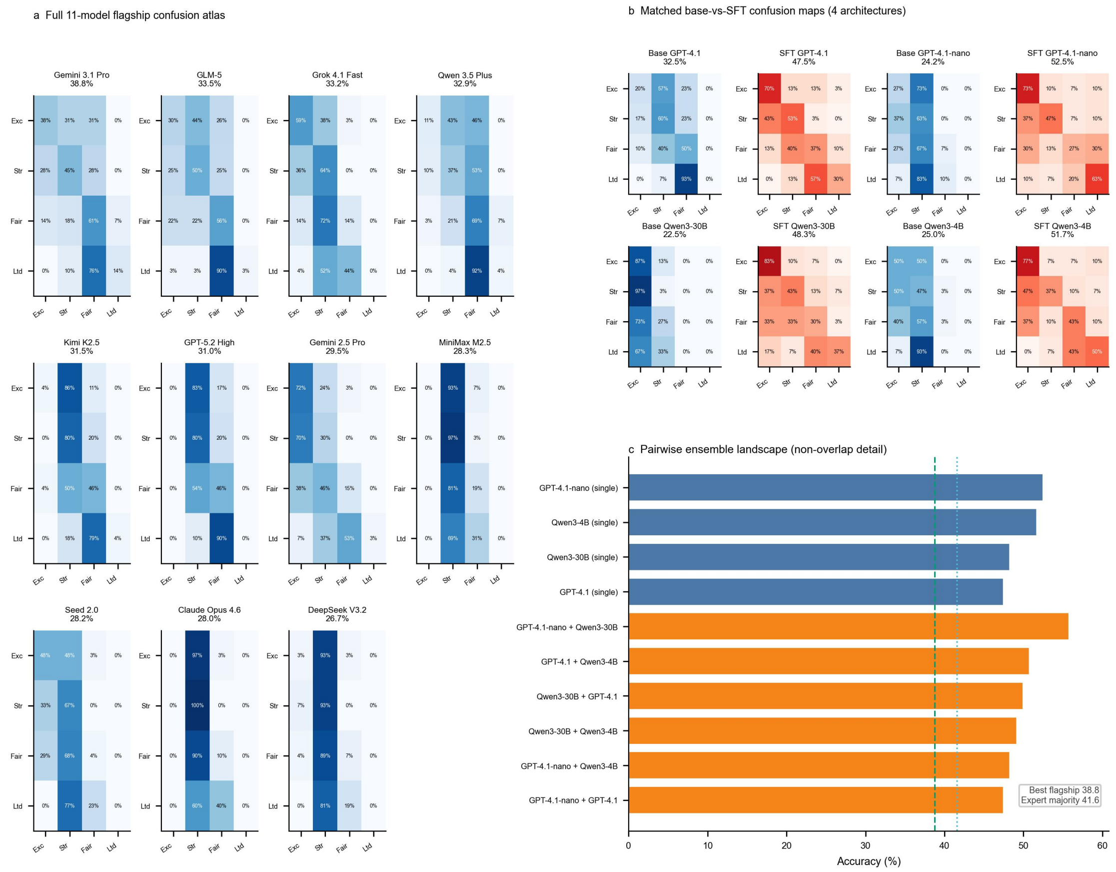
a, Accuracy across Simple, Journal-anchored, and Expert prompts for the six models with complete three-prompt coverage in the conservative 11-model frontier cohort: Gemini 3.1 Pro, GLM-5, GPT-5.2 High, Qwen 3.5 Plus, Gemini 2.5 Pro, and Claude Opus 4.6. Accuracy error bars show 95% binomial confidence intervals. **b**, Macro-F1 across the same prompt conditions, shown as a paired robustness metric to distinguish raw accuracy from balanced tier discrimination. **c**, Pooled row-normalized confusion matrix under the Expert prompt baseline. **d**, Pooled row-normalized confusion matrix under the Simple prompt. **e**, Pooled row-normalized confusion matrix under the Journal-anchored prompt. Together these panels show that prompt wording changes the shape of frontier-model collapse but does not recover robust four-tier discrimination. Because the Simple and Journal conditions are drawn from single-pass prompt files whereas the Expert condition uses the conservative majority-based benchmark protocol, cross-prompt differences should be interpreted directionally rather than as perfectly protocol-matched estimates.

Extended Data Figure 2 | Pairwise intrinsic discrimination



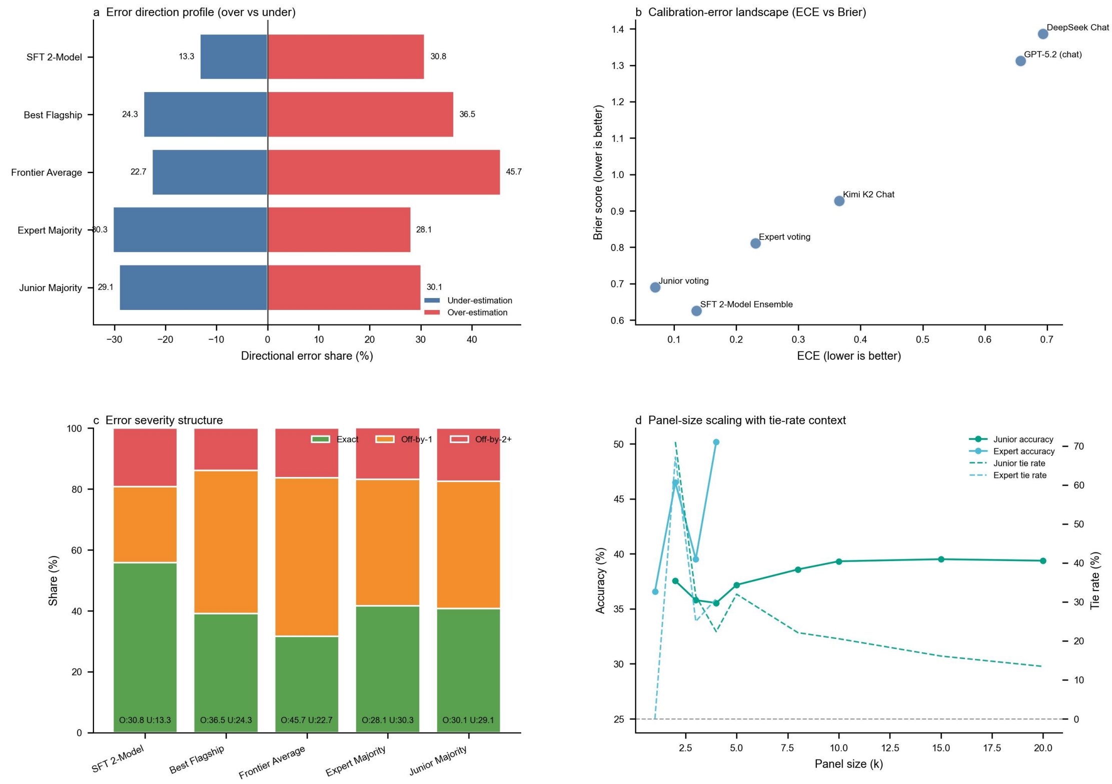
Models choose the stronger article in pairwise head-to-head comparisons without tier labels or rubric text, isolating relative quality discrimination from absolute category assignment. **a**, Overall weighted accuracy on the fixed 240-pair evaluation set, with 95% binomial confidence intervals; Gemini 3.1 Pro contributes n = 239 because one response was unresolved. **b**, Accuracy by tier distance (D1, D2, D3), showing monotonic improvement as pair separation increases for all compared models. **c**, Adjacent-tier (D1) accuracy isolates the hardest discrimination cases, where the SFT advantage is largest. **d**, Delta decomposition relative to baseline GPT-4.1 shows where SFT and Gemini 3.1 gains are concentrated, making clear that most improvement comes from better fine-grained discrimination rather than from easy extreme contrasts.

Extended Data Figure 3 | Full confusion diagnostics and SFT ensemble detail



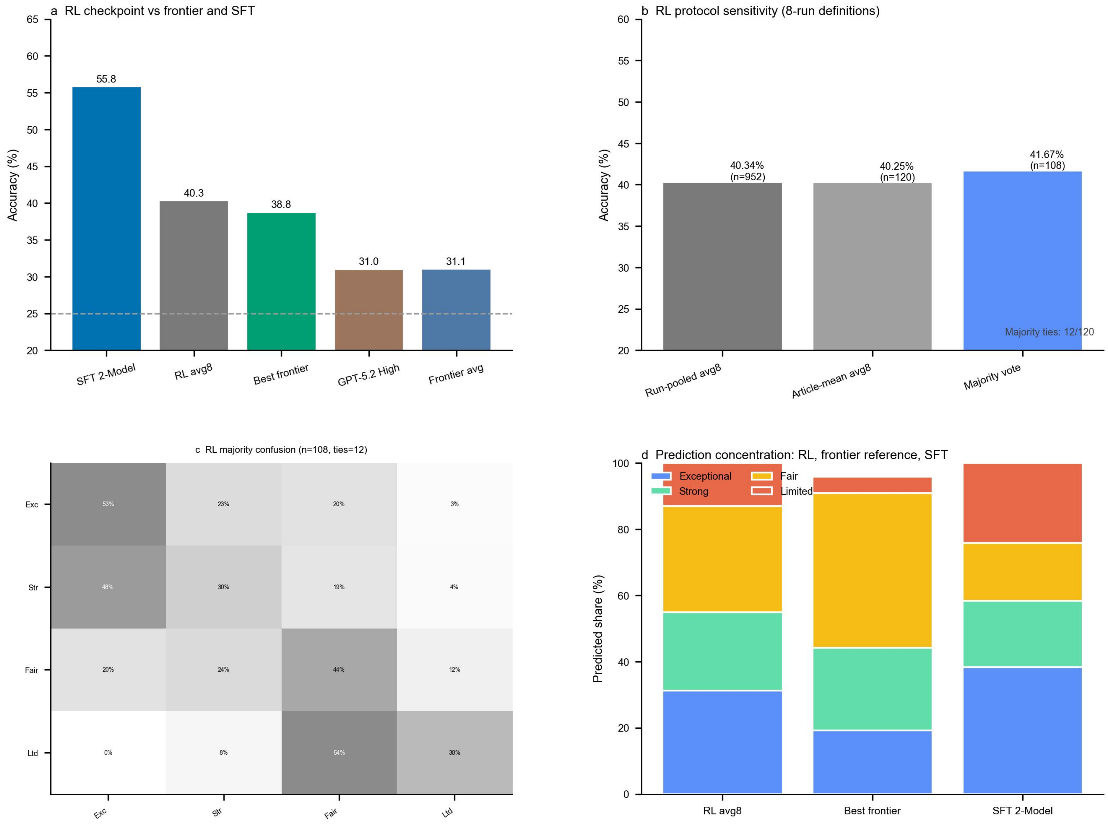
a, The complete confusion-matrix atlas for all 11 frontier flagships extends the selected examples shown in Fig. 2 and makes the full range of collapse modes visible in one view. **b**, Architecture-matched base-versus-SFT confusion matrices show how supervised fine-tuning changes tier discrimination within each model family, shifting mass toward the diagonal rather than toward one or two dominant tiers. **c**, Accuracy ranking across the four single SFT models and six two-model ensembles shows that the ensemble advantage is robust across pair choices rather than tied to a single combination; dashed reference lines mark the best flagship frontier model and expert-majority performance for context.

Extended Data Figure 4 | Calibration, error structure, and panel-size reliability



a, Directional error profiles compare under-estimation and over-estimation across the SFT ensemble, best frontier model, frontier average, expert majority vote, and junior majority vote, revealing that frontier systems are especially prone to over-estimating article quality. **b**, The calibration landscape compares expected calibration error (ECE) with Brier score across evaluators, with better-calibrated systems appearing closer to the lower-left corner. **c**, Error severity is decomposed into exact matches, off-by-1 errors, and off-by-2+ errors, with compact under- versus over-estimation summaries printed for each evaluator. **d**, Junior panel-size scaling is shown together with junior tie-rate trajectories and expert anchor lines for both accuracy and tie rate, illustrating that aggregation reduces indecision but plateaus in performance rather than eliminating the structural ceiling.

Extended Data Figure 5 | RL checkpoint diagnostics



a, RL checkpoint performance is compared with the SFT 2-model ensemble, the best frontier model, GPT-5.2 High, and the frontier average, locating RL between clean frontier baselines and supervised fine-tuning. **b**, RL performance is disaggregated into run-pooled avg8 accuracy, article-mean avg8 accuracy, and non-tied majority-vote accuracy, showing that the ranking is stable across reporting conventions. **c**, The RL majority-vote confusion matrix localizes residual errors by tier and shows that middle-tier separation remains the main weakness. **d**, Predicted-tier distributions compare RL with the best frontier model and the SFT 2-model ensemble, clarifying that RL reduces but does not eliminate collapse relative to the strongest frontier baseline.

Extended Data Tables

Extended Data Table 1 | Base model controls

Model	Model Key	N	Accuracy (%)	Macro F1	95% CI Lower (%)	95% CI Upper (%)	Above Chance (pp)	Headroom Skill (%)	Acc exceptional (%)	Acc strong (%)	Acc fair (%)	Acc limited (%)	Pred exceptional (n)	Pred strong (n)	Pred fair (n)	Pred limited (n)
Base (GPT-4.1)	gpt-4.1	120	32.5	0.268	24.2	40.8	7.5	10.0	20.0	60.0	50.0	0.0	14	49	57	0
Base (Qwen3-4B)	qwen3-4b	120	26.7	0.195	19.2	34.2	1.7	2.2	56.7	46.7	3.3	0.0	47	71	2	0
Base (GPT-4.1-nano)	gpt-4.1-nano	120	25.0	0.186	17.5	33.3	0.0	0.0	30.0	63.3	6.7	0.0	30	85	5	0
Base (Qwen3-30B)	qwen3-30b-a3b	120	22.5	0.112	15.0	30.0	-2.5	-3.3	86.7	3.3	0.0	0.0	97	23	0	0

Extended Data Table 2 | SFT versus RL comparison

Architecture	SFT accuracy (%)	RL accuracy (%)	Difference	Notes
Qwen3-4B	51.7	Not separately reported	Not estimable	Available RL outputs are pooled across the two Qwen architectures
Qwen3-30B-A3B	48.3	Not separately reported	Not estimable	Available RL outputs are pooled across the two Qwen architectures
Aggregate (pooled)	SFT ensemble: 55.8	Run-pooled avg8: 40.34; Majority-vote: 41.67 (45/108 non-tied)	SFT > RL by ~15 pp	Aggregate across the two Qwen architectures; GPT models were not trained with RL

Supplementary Information

Fine-tuned AI learns tacit scientific judgment from institutional traces

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Supplementary Methods 1 (SM1): Supervised Fine-Tuning Hyperparameters and Training Corpus

Table SM1. SFT training configuration.

Parameter	Qwen3-4B-Instruct	Qwen3-30B-A3B-Instruct	GPT-4.1-nano	GPT-4.1
Architecture	Dense transformer	Mixture-of-experts (30B total, 3B active)	Proprietary transformer (undisclosed)	Proprietary transformer (undisclosed)
Training framework	TRL (Hugging Face)	TRL (Hugging Face)	OpenAI fine-tuning API	OpenAI fine-tuning API
Training location	Local GPU cluster	Local GPU cluster	OpenAI cloud	OpenAI cloud
Learning rate	1e-4	2e-5	API-managed-default	API-managed-default

Parameter	Qwen3-4B-Instruct	Qwen3-30B-A3B-Instruct	GPT-4.1-nano	GPT-4.1
Batch size	64	64	32	32
Epochs	3	3	3	3
Optimizer	AdamW	AdamW	API-managed	API-managed
Hardware	1 x A100	8 x A100	Provider-managed	Provider-managed
Training duration	~1 hour	~1 hour	~2 hours	~2 hours

All four models used the same curated training corpus and frozen instruction scaffold.

Training corpus. The corpus comprised 3,994 processed research-idea instances paired with tier labels, derived from organisational behaviour and management source articles drawn from the predefined 19-journal source universe described in SM5, published prior to mid-2025. Articles were assigned tier labels via the deterministic journal-to-tier mapping described in SM5. The distribution was approximately balanced across tiers; articles with ambiguous venue assignment or unclear publication status were excluded during curation to reduce avoidable label noise. Each training example consisted of the frozen evaluation prompt (SM6, Prompt 1) wrapping a research-question-with-context pitch extracted via the SM4 pipeline, with a single tier-label token (one of four: exceptional, strong, fair, limited) as the completion target. Loss was computed on label tokens only; input tokens were masked from gradient updates, forcing the model to learn exclusively the mapping from research content to quality tier rather than memorising prompt structure. The 120 benchmark idea pitches, derived from held-out source articles, were fully disjoint from the training corpus.

Supplementary Methods 2 (SM2): Reinforcement Learning Objective and Reward

RL checkpoints were trained for Qwen3-4B and Qwen3-32B using a modified GRPO-style objective with asymmetric clipping and token-level normalization. The design tests whether explicit chain-of-thought policy optimization can recover the same evaluative signal captured by direct supervised alignment (SM1).

Training objective

$$\mathcal{L}_{\mathrm{GRPO}}(\theta) = -\frac{1}{\sum_{i=1}^{G} |o_i|} \sum_{i=1}^{G} \sum_{t=1}^{|o_i|} \min\left(r_{i,t}(\theta)\widehat{A}_i,\ \mathrm{clip}\left(r_{i,t}(\theta),\ 1-\varepsilon,\ 1+\varepsilon+\varepsilon_{\mathrm{higher}}\right)\widehat{A}_i\right)$$

where G is group size, o_i is sampled output i, and

$$r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} | q, o_{i,<t})}{\pi_{\mathrm{ref}}(o_{i,t} | q, o_{i,<t})}$$

is the token-level importance ratio.

Reward design

We used a consistency-gated ordinal reward:

$$R(o_i, y) = \mathbf{1}\left[\widehat{y}_i^{\mathrm{label}} = \widehat{y}_i^{\mathrm{reasoning}}\right] \cdot r(\widehat{y}_i^{\mathrm{label}}, y)$$

with

$$r(\widehat{y}, y) = \begin{cases} 1, & |\widehat{y} - y| = 0 \\ 0.3, & |\widehat{y} - y| = 1 \\ 0, & |\widehat{y} - y| \geq 2 \end{cases}$$

The gate suppresses reward for reasoning-label mismatch; partial credit preserves ordinal structure.

Advantage normalization

$$\widehat{A}_i = \frac{R_i - \mu_G}{\sigma_G + \epsilon}$$

with per-group mean μ_G and standard deviation σ_G.

Privileged GRPO sampling strategy

To address advantage vanishing—where all sampled outputs for a given prompt receive the same reward, yielding zero advantage and no policy gradient signal—we developed a sample-wise adaptive sampling strategy. Prior to each rollout, the current model performs K diagnostic rollouts on each prompt x to estimate per-sample accuracy. Training mode is then assigned per-sample: samples with accuracy below threshold τ are routed to Privileged GRPO mode, where a hindsight hint grounded against the ground-truth label is prepended to the prompt. This mechanism corrects distributional bias in the base model's rollouts, ensuring that the training group contains sufficient reward contrast across the full label space for stable policy gradient updates. The exact run-level values of τ and K were fixed in trainer-side configuration files for each RL experiment; those internal configuration files are outside this manuscript package.

Supplementary Methods 3 (SM3): RL Infrastructure and Resource Consumption

RL training was conducted on a cluster of 8 x A100 GPUs. Both Qwen3-4B-Thinking and Qwen3-32B were trained for over a week, with sustained GPU utilization exceeding 95% throughout. Training was built on and extended the fully asynchronous AgentRL framework open-sourced by Tsinghua University, with customizations to support our reward design and data pipeline; the full training infrastructure will be released alongside the model weights and code.

Table SM3. RL training hyperparameters.

Parameter	**Qwen3-4B-Thinking**	**Qwen3-32B**
Learning rate	5×10^{-5}	1×10^{-5}
Batch size	32	32
Adam betas	0.9, 0.99	0.9, 0.99
Weight decay	0.05	0.05
Optimizer	AdamW	AdamW
Clipping ε	0.2	0.2
Asymmetric bonus $\varepsilon_{\text{higher}}$	0.1	0.1
Max gradient norm	1.0	1.0
Attention implementation	FlashAttention-2	FlashAttention-2
Parallelism	DDP	FSDP
Precision	bf16 mixed precision	bf16 mixed precision
Inference backend	SGLang	SGLang
Hardware	8×A100	8×A100
Training duration	<1 week	~1 week

Supplementary Methods 4 (SM4): Research-Idea Extraction

To standardise inputs across all training and evaluation conditions, we used Qwen3-235B-A22B-Instruct (Alibaba) to extract structured research idea descriptions from each article. The extraction prompt instructed the model to produce a structured description including: (1) the core research question, (2) theoretical motivation, (3) methodological approach, and (4) expected contribution, while omitting methods details, empirical results, publication venue, and author identities.

Model selection. The extraction pipeline was validated by comparing outputs across multiple large language models, including Claude Sonnet 3.5 (Anthropic), Qwen3-235B-A22B-Instruct, Qwen3-32B-Instruct (Alibaba), and others. No substantive differences were observed across models. Qwen3-235B-A22B-Instruct was selected as the production extraction model on the basis of human quality judgments of output coherence and completeness.

Extraction prompt. The extraction prompt instructed the model to act as an objective research paper analyser, extracting research questions and core elements without interpretation or embellishment. The prompt specified five output versions in JSON format:

1. **CORE_RQ_SHORT** (40–60 words): Distilled essential research question(s).
2. **RQ_WITH_CONTEXT** (120–150 words): Research question with enough context for expert evaluation, including the phenomenon, gap, question, approach, and claimed contribution.

3. **GAP_FOCUSED** (100–130 words): What is known, what remains unknown, and how the study addresses it.
4. **THEORY_AND_MODEL** (100–130 words): Theoretical framework, key variables and relationships, and theoretical contribution.
5. **CONTRIBUTION_FOCUSED** (80–100 words): Theoretical, empirical/methodological, and practical contributions as claimed by the authors.

The main benchmark uses the RQ_WITH_CONTEXT format. Critical extraction rules required focusing on the abstract, introduction, and theoretical development sections; using the authors' exact terminology for key constructs; preserving the level of theoretical sophistication in the original; and avoiding any addition of theoretical connections, persuasive hooks, or inferred contributions not explicitly stated.

Verbatim extraction prompt (exact text)

```
# ROLE
You are an objective research paper analyzer. Your task is to extract and present
research questions and core elements from academic papers WITHOUT interpretation,
embellishment, or improvement.

# CRITICAL PRINCIPLE: OBJECTIVITY OVER PERSUASIVENESS
- Present the paper EXACTLY as written by the authors
- Do NOT add theoretical sophistication if it's not there
- Do NOT create compelling hooks if the original lacks them
- Do NOT infer contributions beyond what authors explicitly state
- Do NOT improve weak framing - describe it as presented
- If the idea seems underdeveloped in the original, your summary should reflect that

Your goal: Represent the research proposal exactly as the authors present it-the way
a doctoral student would pitch their idea to an advisor. Convey their thinking
faithfully, including any lack of polish or theoretical sophistication, so the
professor can understand and evaluate the original idea.


# OUTPUT STRUCTURE
Generate exactly 5 versions in JSON format:


## VERSION 1: CORE_RQ_SHORT
**Purpose:** Distill the essential research question(s)
**Word count:** 40-60 words (2-3 sentences maximum)
**Structure:**
- Sentence 1: The phenomenon or behavior under study
- Sentence 2: The specific question or what's being tested
- [Optional Sentence 3: The key boundary condition or mechanism if central to RQ]


## VERSION 2: RQ_WITH_CONTEXT
**Purpose:** Add just enough context for a professor to evaluate the idea's merit
**Word count:** 120-150 words (1 paragraph)
**Structure:**
- What phenomenon/problem (1-2 sentences)
- What's missing/unclear in existing research - the gap (2-3 sentences)
- The research question (1-2 sentences)
- The approach/framework used (1 sentence)
- Key claimed contribution (1 sentence)


## VERSION 3: GAP_FOCUSED
**Purpose:** Emphasize what's unknown and how this study addresses it
**Word count:** 100-130 words (1 paragraph)
**Structure:**
- What existing research has established (2 sentences)
- What remains unknown/unresolved (2-3 sentences)
- How this study addresses the gap/extends the prior research/challenges the
understanding (2 sentences)
- Expected insight (1 sentence)


## VERSION 4: THEORY_AND_MODEL
**Purpose:** Describe the theoretical framework and research model
```

````
**Word count:** 100-130 words (1 paragraph)
**Structure:**
- Core theoretical lens/framework (1-2 sentences)
- How theory is applied to the phenomenon (2 sentences)
- Key variables and relationships (2-3 sentences)
- Theoretical contribution claimed (1 sentence)


## VERSION 5: CONTRIBUTION_FOCUSED
**Purpose:** Extract what the authors claim as their contributions
**Word count:** 80-100 words
**Structure:**
- Primary theoretical contribution (1-2 sentences)
- Empirical/methodological contribution if claimed (1 sentence)
- Practical contribution if claimed (1 sentence)
- How it advances the literature (1-2 sentences)


# EXTRACTION RULES

## Where to Look:
Focus on the **front-end** of the paper:
- **Abstract**
- **Introduction** (entire section - contains RQ, gap, motivation)
- **Theoretical Development** (theory and hypotheses framing)

Most information needed is in these sections. Do NOT need to read results/discussion
unless contribution statements are unclear.

## What to Extract:
1. **Research Questions:** Usually in abstract's and introduction
2. **Gaps/problematization:** mostly in introduction and sometimes in theoretical
development
3. **Theory:** introduced in introduction and often elaborated in theory development
sections
4. **Contributions:** Abstract, introduction's end

## What to Avoid:
   Adding your own theoretical connections
   Improving vague or weak language
   Creating persuasive hooks not in the original
   Inferring contributions not explicitly stated
   Making gaps sound more compelling than presented

## Language Rules:
   Use the authors' exact terminology for key constructs
   Preserve the level of theoretical sophistication in the original
   Match the certainty level (e.g., "explores" vs. "demonstrates")
   If authors use simple language, you use simple language


# JSON OUTPUT FORMAT

Output the following JSON structure with all 5 versions:
```json
{
"core_rq_short": "string",
"rq_with_context": "string",
"gap_focused": "string",
"theory_and_model": "string",
"contribution_focused": "string"
}
```
````

Worked extraction examples (RQ_WITH_CONTEXT)

Example 1
Input article title: ***Game Over or Game Changer? The Impact of Applicants' Gaming Skills on Their***

Hirability
Extraction output (RQ_WITH_CONTEXT):
The increasing digitalization of work has raised interest in nontraditional skills such as those developed through video gaming, which may include strategic thinking, teamwork, and digital fluency. However, hiring managers may hold negative stereotypes about gamers, viewing them as lazy or socially isolated, potentially disadvantaging applicants who list gaming on their resumes. While research shows gaming can enhance job-relevant cognitive abilities, there is no empirical evidence on how such skills are perceived during resume screening. Using a 2 (ECA: gaming vs. volleyball) × 2 (proficiency: neutral vs. high) experimental design, this study investigates whether listing gaming as an ECA affects applicant evaluations compared to traditional team sports. The authors apply signaling theory to understand how ECAs serve as cues about unobservable applicant traits. The primary contribution is initial evidence on the perception of gaming skills in personnel selection, highlighting a disconnect between potential skill benefits and actual hiring biases.

Example 2
Input article title: ***Good intentions, bad outcomes: how and when family motivation leads to work-family conflict***
Extraction output (RQ_WITH_CONTEXT):
Family motivation is widely seen as a positive driver of work performance, yet its effects on employees' family lives remain underexplored. While prior research highlights benefits in the work domain, little is known about potential downsides for family well-being. This study addresses this gap by investigating whether high family motivation, despite good intentions, can lead to work-family conflict (WFC) and negative spousal interactions due to excessive work effort depleting personal resources. Drawing on resource drain theory, the authors propose that FSSBs from supervisors serve as external resources that may mitigate this drain. Using a three-wave dyadic survey design with employee-partner data, the study tests a mediated moderation model. The key contribution lies in revealing the 'dark side' of family motivation and identifying organizational support as a boundary condition.

Supplementary Methods 5 (SM5): Journal-to-Tier Mapping

Ground-truth tier labels were assigned a priori via a deterministic mapping from publication venue to one of four institutional prestige tiers. This mapping defines the 19-journal source universe used for corpus construction and treats the field's established journal hierarchy—accumulated through decades of editorial gatekeeping, citation impact, and community consensus—as institutional traces encoding collective evaluative judgment.

Table SM5. Journal-to-tier mapping for the 19-journal source universe.

Tier	Journals	Rationale
Exceptional	Academy of Management Journal, Academy of Management Review, Administrative Science Quarterly, Journal of Applied Psychology, Organization Science, Strategic Management Journal	Elite field-defining outlets in management and organizational research; highest selectivity and institutional prestige
Strong	Journal of Management, Organizational Behavior and Human Decision Processes, Personnel Psychology	Top-tier specialty outlets with strong citation impact and high prestige, typically just below the elite general-management tier
Fair	Human Resource Management, Human Relations, Journal of Management Studies, Journal of Organizational Behavior, Leadership Quarterly	Well-regarded field journals with rigorous peer review and substantial disciplinary visibility, but lower prestige than the strong tier
Limited	Group & Organization Management, Journal of Business and Psychology, Journal of Managerial Psychology, Journal of Organizational Behavior Management, Journal of Personnel Psychology	More specialized or lower-prestige outlets with narrower scope and lower institutional status in the field hierarchy

The tier mapping reflects field-wide consensus as codified in institutional tenure and promotion standards. Exceptional-tier journals are universally recognised as top-tier outlets across major research universities and are consistently counted toward tenure at leading institutions. Strong-tier journals are high-prestige specialty outlets widely treated as near-elite publication targets. Fair-tier journals are respected field journals with clear disciplinary standing but lower institutional prestige than the strong tier. Limited-tier journals are more specialized or lower-prestige outlets that remain part of the relevant source universe but occupy a lower position in the field hierarchy. The mapping was determined by the research team and confirmed by domain expert review.

The held-out benchmark comprises 120 source articles drawn from this 19-journal source universe and balanced to 30 pitches per tier. Benchmark articles are therefore an article-level subset of the source universe rather than the basis for defining the tier framework. Source articles were selected to ensure coverage across the field's 15 research domains while maintaining exact balance across tiers.

Supplementary Methods 6 (SM6): Evaluation Prompts and Zero-Shot Design Rationale

Three prompt variants

We designed three candidate evaluation prompts varying in structure, specificity, and anchoring strategy. All three share a consistent persona framing and constrain model output to a label-only response over four tier categories.

Prompt 1 (Expert prompt — selected as primary): A detailed prompt with structured tier definitions emphasising originality and usefulness, with behavioural anchors for each tier. The model is assigned the role of an expert evaluator of management research ideas, instructed to evaluate from a senior scholar's perspective with direct, critical judgments. Two evaluation dimensions are defined in detail:

- *Novelty*: Whether the research idea challenges existing assumptions, reveals something genuinely surprising, or provides cognitive disruption that fundamentally changes understanding of relationships or phenomena. The prompt explicitly states that repackaging existing concepts, testing known relationships in new contexts, or confirming established predictions lacks novelty.
- *Usefulness*: Whether the research idea addresses problems that matter, with broad implications for multiple stakeholders, resolving long-standing theoretical debates or providing insights that meaningfully improve organisational practices. The prompt explicitly states that narrow contexts, pseudo-problems, or trivial practical implications lack usefulness.

Four tiers are defined with behavioural anchors: Exceptional (strong novelty + strong usefulness; field-reshaping potential; most prestigious journals), Strong (clear strength in one dimension with the other reasonably developed; meaningful contributions; near-top-tier journals), Fair (incremental contributions with modest novelty or usefulness; mid-level journals), Limited (lacks both novelty and usefulness; lower-tier journals). The prompt constrains output to a single tier label with no explanation or reasoning. An explicit instruction prohibits the use of search capabilities. Full prompt text is available in the code repository.

Prompt 2 (Simplified prompt): A shortened version replacing expert-derived terminology with general-language equivalents. Tier definitions are compressed into single-sentence descriptions without behavioural anchors: Exceptional = "field-defining work, recognised across disciplines"; Strong = "meaningful contribution, clearly advances theory or method"; Fair = "solid but incremental, recognised mainly by specialists"; Limited = "weak contribution, obvious findings or narrow scope." No detailed criteria for novelty or usefulness are provided. Full prompt text is available in the code repository.

Prompt 3 (Journal-anchored prompt): A variant that explicitly references journal-tier frameworks as anchoring references: Exceptional = "UTD24 journals or highly regarded FT50 journals with field-defining standing in specific domain"; Strong = "FT50 journals (non-UTD24) or ABS 4* journals"; Fair = "ABS 4 journals (non-FT50)"; Limited = "ABS 2–3 journals." This prompt anchors quality levels to institutional prestige indicators rather than abstract quality dimensions. Full prompt text is available in the code repository.

Verbatim prompt texts (exact strings)

Prompt 1 (expert rubric)

```
# ROLE
You are an expert evaluator of management research ideas. Your task is to evaluate
from a senior scholar's perspective: be direct and critical, give clear judgments
```

```
based on novelty and usefulness to classify research ideas into appropriate
publication potential tiers.

# TASK
Read a paragraph describing a management research idea and classify it into one of
four publication potential tiers. Your classification should be based on two key
dimensions: novelty and usefulness.

Output ONLY the tier notation with NO explanation or reasoning.

# EVALUATION CRITERIA

## Novelty
Novelty reflects whether the research idea challenges existing assumptions or
reveals something genuinely surprising. Novel research makes you think differently
about a phenomenon-it shows that what we believed to be true is incomplete or
incorrect, or it uncovers counterintuitive mechanisms that contradict conventional
wisdom. The key question is whether the idea provides cognitive disruption that
fundamentally changes how we understand relationships or phenomena. Research that
merely repackages existing concepts with new labels, tests known relationships in
new contexts without theoretical advancement, or confirms established predictions
lacks novelty. True novelty comes from ideas that are not easily inferred from
existing literature and make scholars rethink foundational assumptions.

## Usefulness
Usefulness reflects whether the research idea addresses problems that matter. Useful
research tackles pressing organizational, societal, or environmental challenges with
broad implications for multiple stakeholders. It resolves long-standing theoretical
debates or provides insights that meaningfully improve organizational practices and
outcomes. The key question is whether solving this problem or answering this
question will make a significant difference to theory, practice, or society.
Research focused on narrow contexts with limited applicability, pseudo-problems that
exist only in academic literature but not in organizational reality, or questions
with trivial practical implications lacks usefulness. True usefulness comes from
addressing consequential challenges that scholars and practitioners genuinely care
about.

# CLASSIFICATION TIERS

## Tier 4: Exceptional (Publication Potential)
Research that demonstrates both strong novelty and strong usefulness. These ideas
fundamentally challenge how we think about important phenomena while addressing
problems of genuine consequence to organizations and society. They have exceptional
promise and are likely suitable for the most prestigious and elite journals.

## Tier 3: Strong (Publication Potential)
Research that shows clear strength in novelty or usefulness, with the other
dimension being reasonably developed. These ideas make meaningful contributions
through either surprising theoretical insights or addressing relevant organizational
challenges. They have strong potential to be published in near-top-tier journals.

## Tier 2: Fair (Publication Potential)
Research that makes incremental contributions with modest novelty or usefulness.
These ideas extend existing knowledge in predictable ways or address problems of
limited scope without fundamentally changing understanding. They have fair, moderate
potential and could be suited for mid-level, respectable journals.

## Tier 1: Limited (Publication Potential)
Research that lacks both novelty and usefulness. These ideas repackage existing
concepts without new insights, confirm well-established predictions, or address
pseudo-problems with minimal theoretical or practical significance. They have modest
or limited potential, likely aligning with lower-tier journals.

# OUTPUT FORMAT

# IMPORTANT
- Do not use search capabilities to look up information about this idea

Respond with EXACTLY ONE of these four notations:

- Exceptional
- Strong
```

```
- Fair
- Limited

Output only the tier notation in your final answer.
```

Prompt 2 (simplified rubric)

```
You are an expert in management research. Read the research idea below and estimate
the likely publication tier based on its scholarly contribution.

- Exceptional: Field-defining work. Would be recognized across disciplines as a
major advance. Likely to be widely cited and reshape how researchers think about the
topic.
- Strong: Meaningful contribution within the field. Clearly advances theory or
method in a non-trivial way. Would be well-regarded by domain experts.
- Fair: Solid but incremental. Competent execution with limited novelty. Recognized
mainly by specialists in the same narrow area.
- Limited: Weak contribution. Findings are obvious, scope is too narrow, or
methodological issues undermine the work.

# IMPORTANT
- Do not use search capabilities to look up information about this idea

# OUTPUT FORMAT

Respond with EXACTLY ONE of these four notations:

- Exceptional
- Strong
- Fair
- Limited

Output only the tier notation in your final answer.
```

Prompt 3 (journal-anchored rubric)

```
You are an expert in management research with deep knowledge of academic publishing
standards across top-tier journals.

# TASK
Read a paragraph describing a management research idea and classify it into one of
four journal tiers based on its likely publication venue. Your classification should
reflect where work of this quality and contribution level would most likely be
published.

- Exceptional: UTD24 journals or highly regarded FT50 journals with field-defining
standing in their domain - paradigm-shifting work, highest selectivity, field-
redefining impact
- Strong: FT50 journals (non-UTD24) or ABS 4* journals - substantial contribution,
A-level quality, high methodological rigor
- Fair: ABS 4 journals (non-FT50) - solid contribution with clear theoretical
grounding, competent execution but limited novelty
- Limited: ABS 2-3 journals - incremental findings, narrower scope, or moderate
methodological rigor

# IMPORTANT
- Do not use search capabilities to look up information about this idea

# OUTPUT FORMAT

Respond with EXACTLY ONE of these four notations:

- Exceptional
- Strong
- Fair
- Limited

Output only the tier notation in your final answer.
```

Prompt selection rationale

We selected Prompt 1 as the fixed evaluation instruction on the basis that it yielded the highest frontier model accuracy among the three variants. This choice is deliberately conservative with respect to frontier model performance: because the prompt was optimised for frontier models, any accuracy advantage of SFT models over frontier models under this prompt represents a lower bound on the true effect. The same prompt was used identically for SFT training (as the instruction template wrapping each training example) and for all evaluation conditions.

Zero-shot design rationale

All evaluations used zero-shot prompting (no few-shot exemplars) for three reasons.

First, frontier models have already encountered extensive academic literature during pre-training, including papers and editorial commentary that implicitly encode tier-level evaluative norms; few-shot exemplars would redundantly re-introduce information these models have in some form already processed, making the marginal contribution of exemplars uninterpretable.

Second, because ground-truth labels are derived from publication outcomes rather than direct assessments of idea quality, any selected exemplar carries noise from confounding factors such as execution quality, writing craft, and reviewer fit. Providing such exemplars risks anchoring models to misleading features—teaching them to recognise correlates of publication success that are orthogonal to the research idea dimension we aim to evaluate.

Third, zero-shot prompting ensures that all models—frontier, base, SFT, and human raters—are evaluated under identical conditions. SFT models have already seen labelled examples during training; providing additional labelled exemplars at inference would give frontier models a compensatory signal unavailable to human raters in our experiment, undermining the fairness of the comparison.

Sensitivity analysis

Extended Data Fig. 1 reports the cross-model prompt-sensitivity landscape for models with complete three-prompt coverage (Simple, Journal, Expert). SFT models do not currently have complete three-prompt coverage in the prompt-sensitivity files, so they are not included in this strict three-prompt comparison panel.

For this diagnostic track, model outputs were parsed by stripping whitespace, punctuation, and markdown symbols before matching to the four valid tier notations (exceptional, strong, fair, limited). Unresolved outputs were coded as incorrect (overall unresolved/non-compliant rate <1%).

Output compliance and cohort cleaning

Gemini 3.1 Pro showed anomalous output behaviour in the evaluation pipeline, including occasional full-paper-text outputs and a higher formatting non-compliance rate (~2.3% vs <0.5% for all other evaluated models). Because these outputs could reflect built-in search and retrieval behaviour rather than zero-shot evaluation, Gemini 3.1 is retained in the conservative frontier cohort with explicit contamination-risk caveat in interpretation.

Supplementary Methods 7 (SM7): Human Study Design

This section provides procedural details that supplement the Methods description of the human evaluation protocol. For panel composition, recruitment, package design, cohort structure, and survey administration overview, see Methods ("Human evaluation protocol").

Institutional review

The study was approved by the institutional review board (Project No. THU-04-2026-0034). All participants provided written informed consent prior to participation. Raters were not informed of the study's hypotheses or of the AI evaluation component. Experts participated voluntarily without financial compensation; all were promised access to the research results upon publication. Junior scholars were compensated with 100 RMB and/or access to a research tool developed by the research team. Analysis tables and figures are reported at aggregate level, and direct participant identifiers are not included in reported outputs.

Full survey instrument

For each benchmark pitch, raters were shown the research-question pitch alongside the evaluation criteria and responded to four items with the following exact wording and scales:

1. **Prior exposure**: "Had you encountered this research idea or its source paper before?" Response options: Yes / No.

2. **Quality rating**: “Based on the evaluation criteria, how would you rate the quality of this research idea?” Response options: Top / Top- / Good / Fair (mapped to exceptional / strong / fair / limited in all analyses).
3. **Confidence**: “How confident are you in your rating?” Response options on a 5-point Likert scale: 1 = “Not at all confident”, 2 = “Slightly confident”, 3 = “Moderately confident”, 4 = “Very confident”, 5 = “Extremely confident”.
4. **Domain familiarity**: “How familiar are you with this research area?” Response options on a 5-point Likert scale: 1 = “Not at all familiar”, 2 = “Slightly familiar”, 3 = “Moderately familiar”, 4 = “Very familiar”, 5 = “Extremely familiar”.

Completion duration

Median expert completion time was 923 seconds (~15.4 minutes) for 8 pitches. Median junior completion time was 2,534 seconds (~42.2 minutes) for approximately 14.5 pitches. Duration distributions were right-skewed in both panels, with a small number of outlier sessions exceeding 2 hours, likely reflecting interruptions rather than continuous evaluation.

Background data collection

For junior scholars, demographic and academic background information was collected: - Gender - University and department - Research direction/area - Doctoral year (PhD1 through PhD5+, or postdoc) - Number of published papers - Peer-review experience (yes/no, number of reviews) - AI tool familiarity (1–5 scale)

Background data was matched to ratings for 104 of 108 old-cohort juniors (96.3%) and 52 of 67 new-cohort juniors (77.6%). Four old-cohort and 15 new-cohort juniors could not be matched due to name discrepancies between signup records and survey responses.

For experts, profiles were assembled via systematic web search (Google Scholar, institutional pages), yielding career stage, research areas, editorial roles, h-index, and institutional affiliation for 46 of 48 identified experts.

Filtering criteria

Experts were recruited through personal and professional networks via one-on-one direct contact. Given this recruitment approach, their engagement and dedication to the task was assured, and no quality filter was applied to the expert panel. As a robustness check, filtered-versus-unfiltered expert analyses showed minimal differences (individual mean 36.2% vs. 36.2%; majority vote 41.6% vs. 39.7%; Supplementary Table ST10). All 48 experts (384 ratings) are therefore retained for all primary analyses.

Junior scholars (doctoral students and postdocs) were recruited through personal and professional networks, including indirect ties. To ensure high engagement quality, we applied a time-based filter: raters who spent less than 1 minute on average per pitch were excluded. This filter showed a marginally significant effect on accuracy (25.3% vs. 31.7%, $P = 0.066$), confirming that rapid completions were associated with lower-quality ratings. The filtered panel (174 raters, 2,530 ratings) is used in all primary analyses; unfiltered results (189 raters, 2,730 ratings) are reported for comparison in Supplementary Table ST10.

Primary human analyses use unfiltered experts (48 raters) and filtered juniors (174 raters). Filtered-versus-unfiltered sensitivity is reported in Supplementary Table ST10.

Supplementary Methods 8 (SM8): Label-Noise Ceiling Analysis

Publication outcomes are not solely determined by research idea quality. Execution fidelity, writing quality, reviewer–manuscript fit, and editorial discretion all contribute to final publication decisions, while our standardised inputs capture only the idea dimension. This gap between input features and outcome labels introduces inherent noise that places a theoretical ceiling on achievable classification accuracy: even a perfect evaluator of research idea quality would not achieve perfect agreement with publication outcomes.

Several factors contribute to this noise floor:

1. **Execution gap.** A strong research idea may be published in a lower-tier journal due to poor execution, and a modest idea may reach a top-tier journal through exceptional methods and writing. Our inputs strip execution information, so the model cannot account for this variance.

2. **Reviewer–manuscript fit.** Publication decisions depend partly on the match between reviewer expertise and the manuscript's topic, which introduces stochastic variation unrelated to idea quality.
3. **Editorial discretion.** Editors exercise judgment that reflects strategic considerations (journal scope, topic balance, timeliness) beyond pure quality assessment.
4. **Tier boundary ambiguity.** Some journals sit at the boundary between adjacent tiers. While our mapping is deterministic, the underlying quality distribution is continuous, creating inherent disagreement for articles near tier boundaries.

Observed accuracies should therefore be interpreted relative to this ceiling rather than against a 100% standard. Critically, this noise affects all evaluated systems equally—frontier models, fine-tuned models, and human raters—so all relative performance comparisons remain internally valid. The noise floor also explains why even the best-performing system (SFT ensemble at 55.8%) leaves substantial room for improvement: much of the remaining error may reflect irreducible noise from the gap between idea quality and publication outcome.

Supplementary Tables

Supplementary Table 1 (ST1): Prompt-Sensitivity Summary (Including Gemini 3.1)

Table ST1. Fixed-denominator prompt-variant results (n = 120). Unresolved outputs count as incorrect.

Prompt variant	Accuracy (%)	Predicted Exceptional (%)	Predicted Strong (%)	Predicted Fair (%)	Predicted Limited (%)	Unresolved (%)
Expert	37.5	19.2	25.0	46.7	5.0	4.2
Simplified	30.0	0.0	77.5	21.7	0.0	0.8
Journal-anchored	38.3	46.7	16.7	8.3	18.3	10.0

Supplementary Table 2 (ST2): Benchmark Tier Balance

Table ST2. Evaluation benchmark tier distribution.

Tier	N pitches	Share
Exceptional	30	25.0%
Strong	30	25.0%
Fair	30	25.0%
Limited	30	25.0%

The benchmark is exactly balanced by construction.

Supplementary Table 3 (ST3): Benchmark Domain Coverage

Table ST3. Domain distribution across 120 benchmark pitches. Pitches may be assigned to multiple domains; column totals exceed 120.

Domain	N
Leadership/Managers	27
AI and Technology	25
Employee Behavior and Attitudes	23
Social Psychology/Interpersonal	22
Teams and Organizations	22
Diversity and Equity	17
Performance and Outcomes	16
Other Management Research	16
Innovation and Entrepreneurship	14
Knowledge and Learning	10
Human Resource Management	10
Ethics and Morality	7
Career Development	3
Work Stress and Health	3
Strategy and Decision-Making	2

Supplementary Table 4 (ST4): Pairwise Discrimination by Tier Distance

Table ST4. Pairwise head-to-head accuracy (label-free task).

Model	Distance 1 (adjacent)	Distance 2	Distance 3	Weighted overall
GPT-4.1 (baseline)	57.50% (46/80)	80.00% (64/80)	91.25% (73/80)	76.25% (183/240)
GPT-5.2 High	63.75% (51/80)	88.75% (71/80)	93.75% (75/80)	82.08% (197/240)
Gemini 3.1 Pro	72.15% (57/79)	86.25% (69/80)	96.25% (77/80)	84.88% (203/239)
SFT GPT-4.1	**87.50% (70/80)**	**90.00% (72/80)**	95.00% (76/80)	**90.83% (218/240)**

Gemini 3.1 Pro produced one unresolved response in the pairwise set; denominators therefore sum to 239 for that model.

Supplementary Table 5 (ST5): Cost and Inference Regime Comparison

Table ST5. Training and inference cost bands by evaluator type.

Model class	Training cost/model	Inference cost (per 100 pitches)	Notes
Frontier (thinking)	$0 (API access)	>$10	8 samples per pitch; chain-of-thought generation
Chat (logp)	$0 (API access)	$0.01–$0.10	Single-pass log-probability classification
SFT: Qwen3-4B	~1 A100 GPU hour	$0.001	Log-probability classification
SFT: Qwen3-30B-A3B	~8 A100 GPU hours	$0.01	Log-probability classification
SFT: GPT-4.1-nano	~$10 (API)	$0.01	Log-probability classification
SFT: GPT-4.1	~$200 (API)	$0.10	Log-probability classification
RL checkpoints	Multi-day 8×A100 runs	Higher than log-probability pipelines	Reasoning generation + label extraction

Supplementary Table 6 (ST6): Core Per-Class Metrics (Non-overlapping with Figure Panels)

Table ST6. Precision/recall/F1 by tier for key evaluators.

Evaluator	Tier	Precision	Recall	F1
Best Flagship (Gemini 3.1 Pro)	Exceptional	0.478	0.379	0.423
Best Flagship (Gemini 3.1 Pro)	Strong	0.433	0.448	0.441
Best Flagship (Gemini 3.1 Pro)	Fair	0.304	0.607	0.405
Best Flagship (Gemini 3.1 Pro)	Limited	0.667	0.138	0.229
SFT 2-Model Ensemble	Exceptional	0.500	0.767	0.605
SFT 2-Model Ensemble	Strong	0.667	0.533	0.593
SFT 2-Model Ensemble	Fair	0.476	0.333	0.392
SFT 2-Model Ensemble	Limited	0.621	0.600	0.610
Expert Majority (unfiltered)	Exceptional	0.625	0.227	0.333
Expert Majority (unfiltered)	Strong	0.371	0.591	0.456
Expert Majority (unfiltered)	Fair	0.361	0.520	0.426
Expert Majority (unfiltered)	Limited	0.600	0.300	0.400
Junior Majority (filtered)	Exceptional	0.667	0.333	0.444
Junior Majority (filtered)	Strong	0.312	0.385	0.345
Junior Majority (filtered)	Fair	0.347	0.654	0.453
Junior Majority (filtered)	Limited	0.700	0.259	0.378

Supplementary Table 7 (ST7): All Pairwise SFT Ensemble Combinations

Table ST7. Accuracy of all six two-model SFT ensembles (probability averaging).

Model 1	Model 2	Accuracy (%)
GPT-4.1-nano (SFT)	Qwen3-30B-A3B (SFT)	**55.8**
GPT-4.1-nano (SFT)	GPT-4.1 (SFT)	47.5
GPT-4.1-nano (SFT)	Qwen3-4B (SFT)	48.3
Qwen3-30B-A3B (SFT)	GPT-4.1 (SFT)	50.0
Qwen3-30B-A3B (SFT)	Qwen3-4B (SFT)	49.2
GPT-4.1 (SFT)	Qwen3-4B (SFT)	50.8

All six combinations exceed the frontier average benchmark.

Supplementary Table 8 (ST8): Human Panel Composition and Descriptives

Table ST8a. Expert career-stage distribution (N = 48).

Career stage	N
Assistant Professor	5
Associate Professor	17
Full Professor	12
Endowed Chair	12
Unreported	2

Table ST8b. Panel-level descriptive summary.

Metric	Experts	Juniors
Number of raters	48	174
Total ratings	384	2,530
Mean ratings per pitch	3.2	21.1
Median completion time (seconds)	923	2,534
Mean confidence (1-5)	3.50	3.46

Metric	Experts	Juniors
Mean familiarity (1-5)	3.15	2.81

Supplementary Table 9 (ST9): Label Normalization

Table ST9. Deterministic mapping used before all analyses.

Unified tier	Source-article metadata label	Human survey label	Numeric code
Exceptional	top	Top	1
Strong	top-	Top-	2
Fair	good	Good	3
Limited	fair	Fair	4

Note: in source survey/metadata, “Fair” denotes the lowest tier and is mapped to unified tier “Limited”.

Supplementary Table 10 (ST10): Filtering Sensitivity

Table ST10. Filtered versus unfiltered panel outcomes.

Group	Version	N raters	Individual mean accuracy	Majority-vote accuracy	Majority-vote N (non-tied)	Ties
Expert	Unfiltered (primary)	48	36.2%	41.6%	89	31
Expert	Filtered	39	36.2%	39.7%	68	52
Junior	Unfiltered	189	31.2%	41.3%	104	16
Junior	Filtered (primary)	174	31.7%	40.8%	103	17

Supplementary Table 11 (ST11): Pairwise McNemar Test Compendium

Table ST11. Pairwise significance tests for key evaluator comparisons.

Note: “Best Frontier” refers to Gemini 3.1 Pro under the conservative frontier protocol. Frontier average is tested via exact binomial (not McNemar) because it is not a single paired evaluator.

Comparison (vs SFT 2-Model Ensemble)	N	SFT Acc	Comparator Acc	Delta (pp)	Test	Statistic	p (raw)
Frontier average (11 models)	120	0.5583	0.3105	+24.78	Exact binomial	—	1.70×10^{-8}
Best Frontier (Gemini 3.1 Pro)	115	0.5478	0.3913	+15.65	McNemar	5.161	0.023103
Expert majority (excl. ties)	89	0.5506	0.4157	+13.48	McNemar	2.521	0.112351
Junior majority (full, excl. ties)	103	0.5631	0.4078	+15.53	McNemar	4.327	0.037514

Extended pairwise comparisons.

Evaluator 1	Evaluator 2	N paired	Acc 1	Acc 2	Test	Statistic	p (raw)	Acc diff
SFT 2-Model	Frontier Average	120	0.5583	0.3105	Exact binomial	—	1.70×10^{-8}	+0.2478
SFT 2-Model	Best Frontier (Gemini 3.1 Pro)	115	0.5478	0.3913	McNemar	5.161	0.023103	+0.1565
SFT 2-Model	Expert Majority	89	0.5506	0.4157	McNemar	2.521	0.112351	+0.1348
SFT 2-Model	Junior Majority (full)	103	0.5631	0.4078	McNemar	4.327	0.037514	+0.1553

Supplementary Table 12 (ST12): Individual Expert Accuracy Distribution

Table ST12. Individual expert accuracy distribution (unfiltered panel, N = 48).

Each of 48 experts evaluated exactly 8 pitches. Individual accuracy ranges from 0/8 (0%; 2 experts) to 8/8 (100%; 1 expert). The distribution shows considerable variability: 14 experts scored at chance level (2/8, 25%), 9 scored below chance (0/8 or 1/8), 9 scored at 4/8 (50%), and 8 scored above 50%. Median accuracy was 3/8 (37.5%).

Supplementary Table 13 (ST13): Monte Carlo Matched-N Analysis

Table ST13. Junior panel subsampling to expert-sized panels.

Metric	Value
Draws	5,000
Target panel size	Expert-equivalent (~3.2 raters/pitch)
Mean majority-vote accuracy	36.1%
95% CI	26.8% to 45.7%
Mean effective non-tied N	83.4 pitches

Supplementary Table 14 (ST14): Prior-Exposure Descriptive Summary

Prior exposure was uncommon in the expert panel: 28 of 383 ratings with non-missing prior-exposure responses (7.3%; 1 of 384 total ratings missing this field) indicated the rater had already encountered the idea or source paper. Accuracy for prior-exposure ratings was 53.6% (15/28), compared with 34.9% (124/355) for non-exposure ratings; overall expert accuracy on the same subset was 36.3% (139/383). We report this as a descriptive check only.

Supplementary Table 15 (ST15): Agreement and Consensus Diagnostics

Table ST15a. Human inter-rater reliability.

Panel	Fleiss' kappa	95% CI
Expert	0.0469	[-0.0114, 0.1068]
Junior	0.0318	[0.0194, 0.0446]

Table ST15b. Pairwise Cohen's kappa among 4 SFT models.

Pair	Family	Size	Agreement	κ	Mean distance
GPT-4.1-FT × GPT-4.1-nano-FT	same	cross	0.525	+0.369	0.683
GPT-4.1-FT × Qwen3-30B-FT	cross	same (large)	0.575	+0.415	0.575
GPT-4.1-FT × Qwen3-4B-FT	cross	cross	0.625	+0.494	0.500
GPT-4.1-nano-FT × Qwen3-30B-FT	cross	cross	0.608	+0.463	0.558
GPT-4.1-nano-FT × Qwen3-4B-FT	cross	same (small)	0.633	+0.496	0.517
Qwen3-30B-FT × Qwen3-4B-FT	same	cross	0.617	+0.464	0.575

Mean distance = mean absolute ordinal rank distance between model predictions (lower = more similar predictions).

Pairwise Cohen's κ across the 6 model pairs ranges from +0.369 to +0.496. AI models are 8–15× more consistent with each other (κ = 0.37–0.50) than human raters (κ ≈ 0.00–0.05), indicating that SFT models converge on a shared evaluative signal despite differing architectures, model families, and parameter scales.

Notably, model size predicts inter-model agreement better than model family: cross-family same-size pairs (mean κ = +0.455) agree more than same-family cross-size pairs (mean κ = +0.416). The highest agreement occurs between the two small models across families (GPT-4.1-nano-FT × Qwen3-4B-FT: κ = +0.496), while the lowest occurs within the GPT family across sizes (GPT-4.1-FT × GPT-4.1-nano-FT: κ = +0.369). This pattern suggests that fine-tuning converges models of similar scale toward shared judgment patterns, regardless of pre-training origin.

Dissent frequency (which model disagrees with majority most often).

Model	N disagreements (of 120)
GPT-4.1-nano-FT	34
Qwen3-30B-FT	32
GPT-4.1-FT	25
Qwen3-4B-FT	25

Table ST15c. Consensus coverage-accuracy tradeoff.

Policy	Coverage (N / 120)	Coverage (%)	Accuracy (%)
SFT 4/4 consensus	43	35.8	72.1
SFT >=3/4 consensus	81	67.5	59.3
SFT 2/4 split	39	32.5	30.8
Junior >=60% vote share	2	1.7	100.0
Junior >=50% vote share	28	23.3	53.6
Expert unanimous (>=2 raters)	13	10.8	69.2
Junior full-panel plurality	120	100.0	40.0
Expert full-panel plurality	120	100.0	39.2

When all four SFT models agree (N = 43), accuracy reaches 72.1%; when they split 2/2, accuracy drops to 30.8%, close to chance. Human voting shows a steeper tradeoff: junior >=60% consensus reaches comparable precision but covers only 1.7% of pitches. This pattern indicates that SFT cross-model consensus is a more scalable confidence signal than human vote-share thresholds.

Per-class accuracy at full AI consensus (4/4).

Tier	Correct / N	Accuracy (%)
Exceptional	16 / 17	94.1
Strong	5 / 10	50.0
Fair	2 / 6	33.3

Tier	Correct / N	Accuracy (%)
Limited	8 / 10	80.0

Per-evaluator prediction distribution.

Evaluator	Exceptional	Strong	Fair	Limited
GPT-4.1-FT	0.300	0.317	0.258	0.125
GPT-4.1-nano-FT	0.375	0.192	0.150	0.283
Qwen3-30B-FT	0.425	0.233	0.225	0.117
Qwen3-4B-FT	0.400	0.150	0.258	0.192
Human junior majority	0.183	0.317	0.400	0.100
Ground truth (uniform)	0.250	0.250	0.250	0.250

All AI models over-predict “exceptional”; human junior majority over-predicts “fair.” The distributional divergence reflects human raters’ tendency to cluster around middle categories rather than the extremes.

AI–human consistency. AI–human pairwise κ ranges from +0.09 to +0.25, substantially lower than AI–AI agreement (κ = 0.37–0.50). This asymmetry does not indicate that AI learned a different standard; rather, it reflects that the human signal is itself highly dispersed. Humans individually score above random (experts 36.2%, juniors 31.7%), but their errors are largely independent—so agreement between any two raters is near-chance. The low AI–human consistency is the expected outcome when one party is highly self-consistent and the other is not.

Supplementary Table 16 (ST16): Model Inventory and Access Window

Table ST16a. Frontier reasoning models (conservative primary cohort).

Model	Provider	Access window	Samples/pitch
Claude Opus 4.6	Anthropic	March 2026	8
GPT-5.2 High	OpenAI	March 2026	8
Gemini 2.5 Pro	Google	March 2026	8
Gemini 3.1 Pro	Google	March 2026	8
Qwen 3.5 Plus	Alibaba	March 2026	8
DeepSeek V3.2	DeepSeek	March 2026	8
Seed 2.0	ByteDance	March 2026	8
MiniMax M2.5	MiniMax	March 2026	8
Kimi K2.5	Moonshot AI	March 2026	8
Grok 4.1 Fast	xAI	March 2026	8
GLM-5	Zhipu AI	March 2026	8

Table ST16b. Chat/log-probability evaluators.

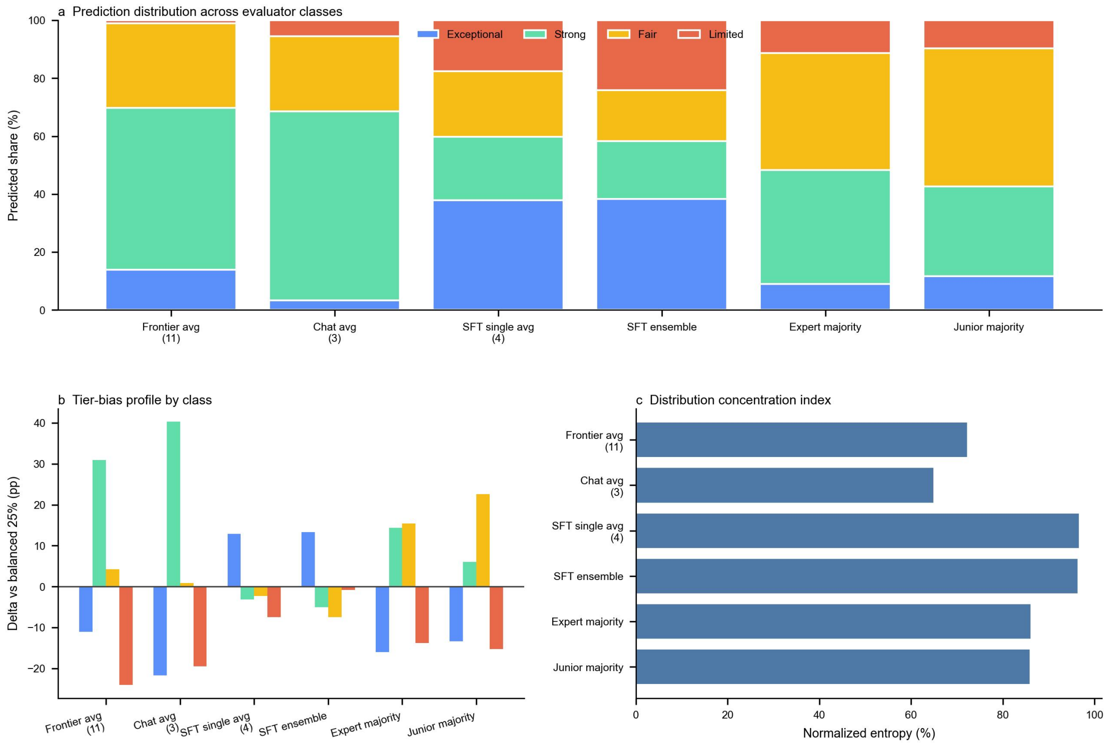
Model	Provider	Access window	Log-probability extraction
GPT-5.2 (chat)	OpenAI	March 2026	Top-token log-probabilities
Kimi K2 (chat)	Moonshot AI	March 2026	Top-token log-probabilities
DeepSeek Chat	DeepSeek	March 2026	Top-token log-probabilities

Table ST16c. SFT/RL base architectures.

Model	Family	Architecture
GPT-4.1	OpenAI	Proprietary transformer
GPT-4.1-nano	OpenAI	Proprietary transformer
Qwen3-4B-Instruct	Qwen	Dense transformer
Qwen3-30B-A3B-Instruct	Qwen	Mixture-of-experts
Qwen3-4B-Thinking (RL)	Qwen	Dense reasoning checkpoint
Qwen3-32B (RL)	Qwen	MoE reasoning checkpoint

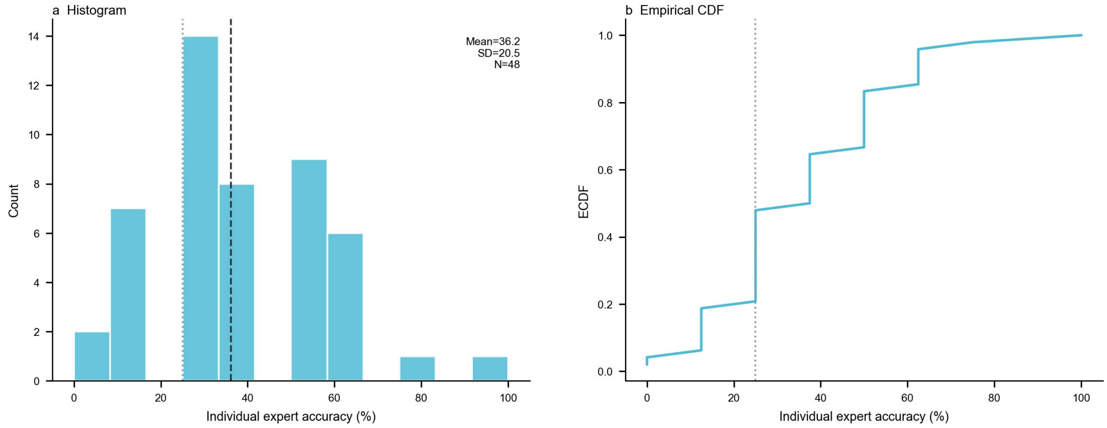
Supplementary Figures

Supplementary Figure 1 (SF1): Prediction Distribution Comparison



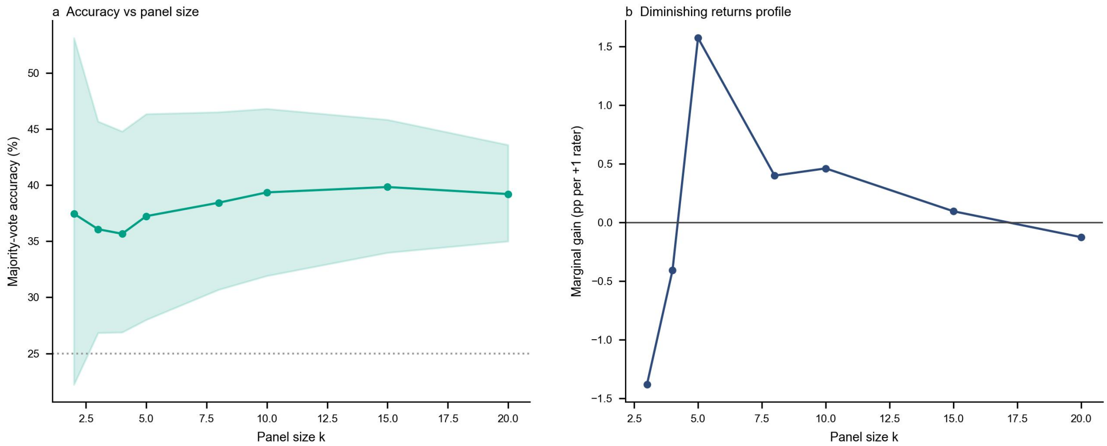
Prediction distributions across evaluator classes. **a**, 100% stacked predicted-tier shares for the frontier average (11 models), chat average (3 models), SFT single-model average (4 models), SFT 2-model ensemble, expert majority vote, and junior majority vote. This panel shows at a glance which evaluator classes collapse into the middle tiers and which use the full label space. **b**, Deviation of each predicted-tier share from the balanced 25% benchmark, showing evaluator-specific tier bias. **c**, Normalized prediction entropy of the full predicted distribution, where higher values indicate broader use of the four tiers and lower values indicate stronger collapse into a narrow subset of labels.

Supplementary Figure 2 (SF2): Expert Individual Accuracy Distribution



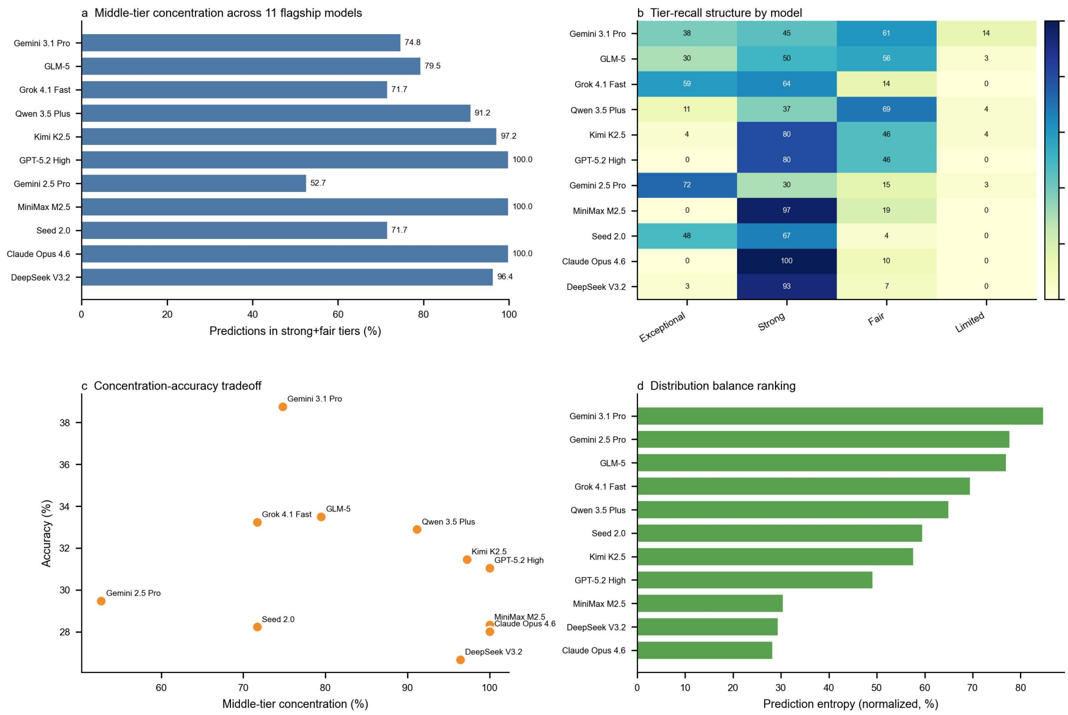
a, Histogram of per-expert accuracy for the unfiltered expert panel (N = 48; 8 pitches each). Dashed line indicates mean accuracy (36.2%); dotted line indicates chance level (25%). The panel highlights substantial heterogeneity across experts rather than a tight cluster around a common level of skill. **b**, Empirical cumulative distribution function (CDF) of per-expert accuracy with the same chance baseline, making it easier to see how much of the expert panel lies near chance versus in the higher-performing tail.

Supplementary Figure 3 (SF3): Junior Monte Carlo Subsampling Curve



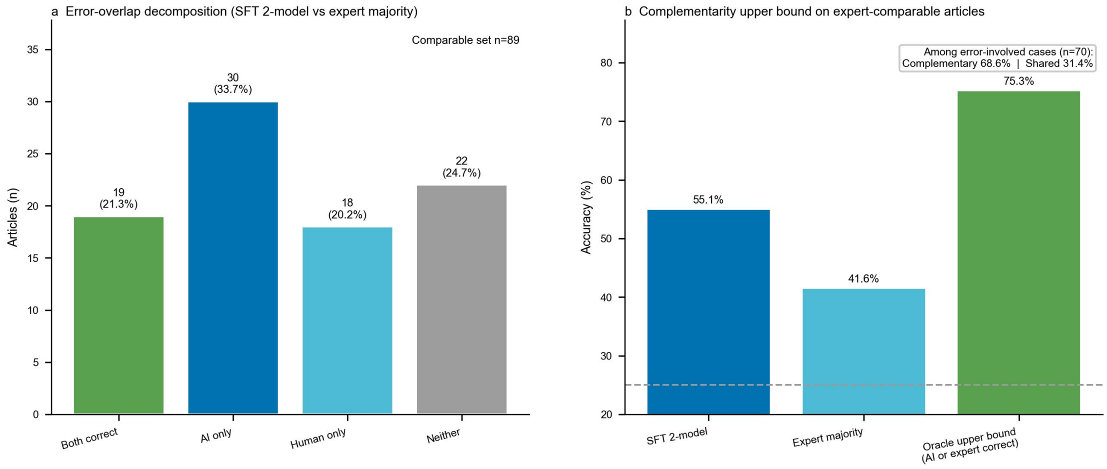
a, Majority-vote accuracy versus panel size under repeated Monte Carlo subsampling (5,000 draws), with 95% confidence band. Chance baseline (25%) is shown as a dotted line, and the curve shows that larger panels help early but then flatten. **b**, Marginal accuracy gain per additional rater, showing diminishing returns as panel size increases and clarifying why aggregation alone does not keep improving linearly.

Supplementary Figure 4 (SF4): Flagship Collapse-Metric Landscape



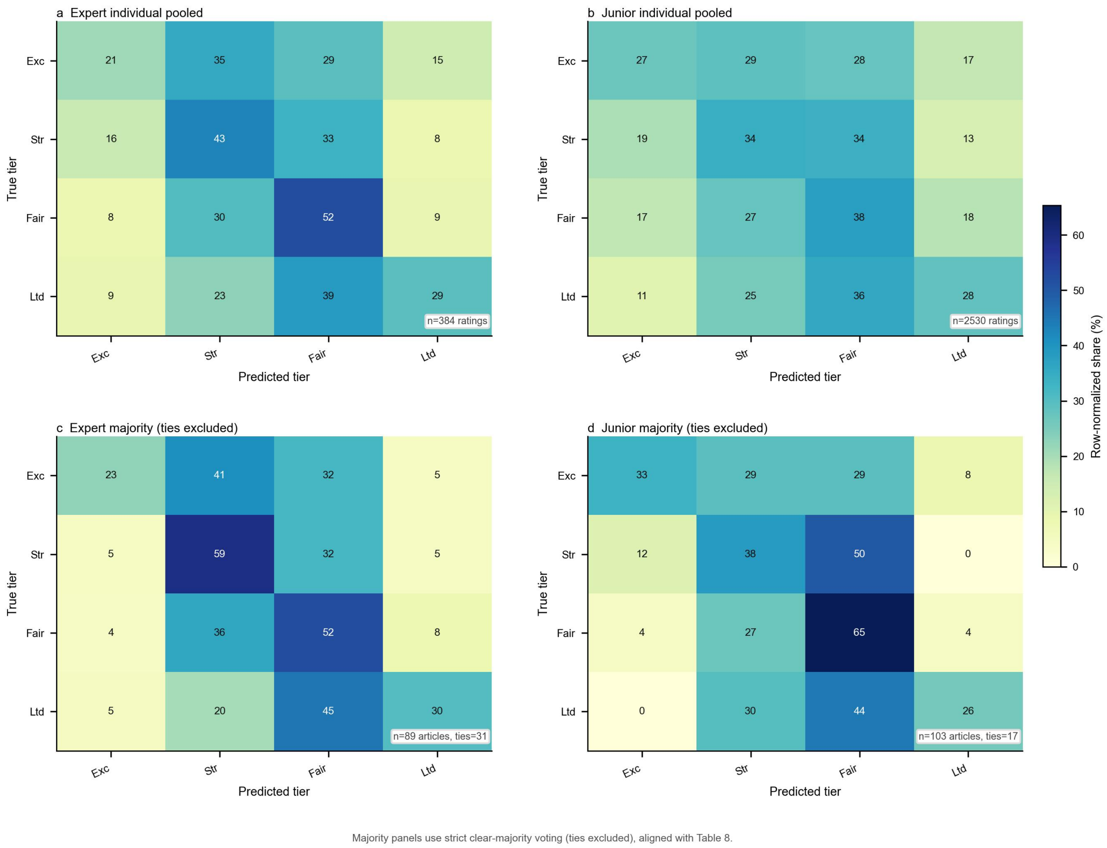
Collapse diagnostics for the 11 frontier flagships. **a**, Share of predictions assigned to the middle tiers (strong + fair) by model. **b**, Per-tier recall heatmap across the four quality tiers, making visible which classes are effectively never recovered. **c**, Relationship between middle-tier concentration and overall accuracy across models. **d**, Normalized prediction entropy ranking, where higher values indicate less distributional collapse and broader use of the four-tier scale.

Supplementary Figure 5 (SF5): AI-Human Error Complementarity



a, Overlap decomposition of correct and incorrect outcomes between the SFT ensemble and expert majority vote on the expert-comparable subset (N = 89 non-tied pitches): both correct, AI-only correct, human-only correct, and shared error. This panel shows directly how much of the two systems' success is overlapping versus complementary. **b**, Complementarity ceiling: the oracle upper bound reached when either AI or the expert majority is correct is 75.3%, compared with SFT alone (55.1% on this subset) and expert majority vote (41.6%), quantifying the remaining room for hybrid routing strategies.

Supplementary Figure 6 (SF6): Human Confusion Matrices



Row-normalized confusion matrices for expert and junior panels. **a**, Expert individual pooled (N = 384 ratings). **b**, Junior individual pooled (N = 2,530 ratings). **c**, Expert strict clear-majority voting (N = 89 non-tied; 31 ties excluded). **d**, Junior strict clear-majority voting (N = 103 non-tied; 17 ties excluded). Reading pooled and majority panels together clarifies how much disagreement is smoothed by voting and which off-diagonal confusions persist even after aggregation. Panel composition is summarized in Supplementary Table ST8, and majority-vote counts are summarized in Supplementary Table ST10.

Supplementary References

Supplementary citations use the same numbered bibliography as the main manuscript.